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Machine-Learned Mismatch and Task Preservation Beliefs in CoSMA DAI for Common Knowledge Aware Semantic Alignment

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Abstract

Correct packet delivery does not guarantee correct semantic interpretation when endpoint meanings for the same learned codeword diverge. CoSMA DAI is proposed for mismatch detection, protected confirmation, task preservation and semantic repair. Channel-conditioned global evidence, semantic class local evidence, temporal dynamics, channel context and protected probe evidence are fused by a causal machine-learned mismatch belief. A transmitter-derived task belief preserves the downstream decision while repair is pending and BDIX agents select guarded intentions for probing, fallback and resynchronisation. Evaluation uses 30 held-out drift seeds, 300 matched null streams and 300 degrading channel controls. Six referenced sequential monitors receive the same conditioned payload score. CoSMA DAI obtains 100.00 percent balanced accuracy, precision, recall, F1 score and Matthews correlation coefficient with zero observed matched null false alarms. Its aggregate delay is 5.62 slots, compared with 11.58 slots for the other zero false alarm method. The task-preservation belief maintains 94.73 percent task accuracy through every divergence scenario, above the quantised accuracy ceiling of 0.919 of the semantic path, because it is derived from the unquantised transmitter latent. A task-label-only control confirms that this accuracy is secured by the preservation belief alone, independently of the detector, so task preservation and mismatch detection are decoupled by design and detectors are compared on residual functional semantic outage, outage duration and semantic reconstruction fidelity, which measure the restoration of the semantic representation itself. Without repair, the residual semantic outage is 73.69 percent at 15.97 dB reconstruction fidelity, whereas CoSMA DAI reduces it to 0.73 percent over 6.62 slots at 21.74 dB. Under five declared parity tiers, in which multivariate and supervised baselines receive the identical features, training seeds, protected probe and candidate budget, the protected confirmation stage reduces false repair for every detector to which it is attached. Zero-shot evaluation over 7 unseen mismatch families and 5 unseen link models retains full detection with zero observed false repair in 6 of the 7 families and on every link and identifies receiver-side decoder drift as a condition the present observation model cannot detect. The learned belief is validated at slot level with an area under the receiver operating characteristic curve of 0.99997 and a class overlap of 0.00039, leave-one-mechanism-out and cross-channel retraining are reported, behaviour is characterised down to the practical detection boundary and scaling to 64-dimensional representations with 2048-entry codebooks is demonstrated. The task-belief mechanism is shown to be economical only for small closed-set output spaces and the channel-conditioning tables are shown to reduce to 6 cells without loss. Every



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comparator is additionally retuned on the same development budget, paired bootstrap intervals and signed-rank tests are reported over the shared streams, auxiliary traffic and radio energy are normalised per correct decision, authentication of the task belief is specified and charged and transfer to MNIST, Fashion-MNIST, CIFAR-10 and CIFAR-100 is demonstrated without retraining, including on a convolutional VQ-VAE representation with a jointly learned 512-entry codebook, where foreground segmentation and localisation are restored to within the quantisation limit while a class decision cannot serve either task. The control traffic share is 23.59 percent, which is 12.62 percent lower than the monitor value. The additional semantic side information increases radio energy to 0.393 mJ per stream and reduces control-adjusted resource efficiency to 6.203 source-equivalent bits per channel use. The results therefore establish reliable detection and semantic repair within the principal comparison, with comparator-specific delay advantages and without claiming task-accuracy, semantic-rate or energy superiority.

Keywords: semantic communication; common knowledge; machine-learned beliefs; task preservation; gradient boosting; distributed artificial intelligence; BDIx agents; semantic codebook alignment; protected probe; explainable control

1. Introduction

Semantic communication is intended to preserve task-relevant meaning rather than every transmitted symbol. Deep semantic transmission has been demonstrated for text through DeepSC and for images through deep joint source and channel coding [1,2]. Digital semantic communication has also been enabled by vector quantised representations in which learned latent vectors are mapped to discrete indices [3,4]. These designs provide substantial communication efficiency, although their correct operation depends on compatible semantic knowledge at the transmitter and receiver.

A semantic index can be delivered correctly by the physical and syntactic protocol while a different meaning is recovered at the receiver. This condition can arise when a codebook, knowledge base, semantic encoder, or task decoder is updated independently. Cooperative learning, split learning, semantic equalisation, knowledge sharing and knowledge distillation have been proposed to reduce this mismatch [5–10]. These approaches mainly address how semantic knowledge should be aligned. A separate question remains concerning when an online repair is justified under channel variation, uncertainty and control cost.

Sequential monitoring can indicate that a statistical score has changed, although the same score movement can be caused by fading, interference, mobility or semantic divergence. Classical cumulative sum and Page-Hinkley procedures provide efficient change detection [11,12]. Anytime valid evidence has been developed through detectors, conformal test martingales, diagnostic martingales, WATCH and conditional conformal test martingales [13–17]. A semantic repair decision must therefore separate channel effects from semantic inconsistency and must confirm the inconsistency before a complete codebook transfer is admitted.

Distributed artificial intelligence provides an appropriate control structure because reasoning can be divided between communication endpoints. Belief, desire and intention agents were formalised by Rao and Georgeff [18] and multiagent learning was reviewed by Stone and Veloso [19]. Distributed artificial intelligence has subsequently been applied to communication control [20,21]. BDIx desire prioritisation and plan selection were examined in [22]. In the present study, the informational belief state is strengthened by a causal

machine learning estimator, while a fixed guarded plan library is retained so that safety conditions, intentions and explanations remain explicit.

The methodological novelty of CoSMA DAI is established by treating semantic alignment as a staged communication and reasoning problem rather than as a single alarm or codebook update. Existing semantic transceivers concentrate primarily on representation quality, existing alignment mechanisms concentrate on knowledge transfer or equalisation and existing sequential monitors concentrate on statistical change. None of these functions alone determines whether an observed inconsistency is semantic rather than channel-induced, whether independent confirmation has been obtained, whether repair is safe under the current link state and how the downstream task should be preserved while confirmation is pending. CoSMA DAI closes this gap by coupling a channel-conditioned evidence process with an active protected probe, a causal mismatch score, an authenticated task preservation belief and an explicit BDIx plan library.

A semantic alarm is not treated as equivalent to a repair command. Passive evidence is first accumulated at global and semantic class levels. A candidate inconsistency then opens a protected diagnostic stage. The independently calibrated probe response is combined with causal payload, temporal and channel observations to revise the semantic mismatch belief. Repair remains blocked until the learned belief, the channel safety condition, the recurrence state and the remaining control budget satisfy the corresponding BDIx guard. During the resulting confirmation interval, a transmitter-derived task decision is retained as compact common knowledge so that a possibly divergent receiver codebook does not immediately determine the application output. The novelty therefore lies not only in using machine learning but also in placing learned beliefs inside an auditable communication protocol whose confirmation, safety, task preservation and resource accounting stages remain explicit.

The principal contributions are summarised as follows.

1. A functional semantic mismatch formulation is introduced for a learned 128-entry digital codebook, coded semantic index delivery and a mobile Rician link with correlated interference. The formulation distinguishes physical packet correctness from semantic meaning consistency.
2. A channel-conditioned evidence mechanism is developed through robust codeword and channel-specific calibration, a persistent global accumulator and semantic class local memories. Broad, gradual and sparse semantic divergence can therefore be represented without allowing ordinary channel variation to dominate the detector state.
3. A causal machine-learned mismatch belief is defined from instantaneous payload evidence, accumulated evidence, temporal dynamics, channel context and protected probe evidence. Future observations and held-out labels are excluded and the resulting score is used as a BDIx belief rather than as an autonomous action.
4. A transmitter-derived task preservation belief is introduced as authenticated semantic side information. The belief temporarily protects the downstream class decision while semantic repair remains pending and its communication cost is separated from probe and resynchronisation control traffic. The belief maintains 94.73 percent task accuracy through every divergence scenario, above the quantised ceiling of the semantic path and an always-on control and equal-resource label-augmented monitors are reported so that this property is attributed to the preservation mechanism rather than to mismatch detection.
5. A complete transmitter and receiver BDIx protocol is specified through acknowledgements, candidate detection, protected probing, belief revision, robust fallback, predictive resynchronisation, full resynchronisation and explanation records. The learned mismatch belief cannot bypass channel, budget, or recurrence guards.

6. An evaluation is conducted against six referenced state-of-the-art sequential monitors using identical conditioned payload inputs, shared held-out streams, matched reliability controls and equal numbers of final configuration candidates. The principal comparison uses different development-data budgets; a separate comparison equalises those budgets. Event detection, delay, task accuracy, overhead, task-weighted data rate, semantic spectral efficiency, control-adjusted resource efficiency, radio energy and control traffic are evaluated jointly.
7. CoSMA DAI obtains 100.00 percent balanced accuracy, precision, recall, F1 score and Matthews correlation coefficient with zero observed matched null false alarms. Relative to the other evaluated zero false alarm method, aggregate delay is reduced by 51.44 percent. Task preservation maintains 94.73 percent task accuracy in every scenario; because a task-label-only control reaches the same value, this is reported as a property of the mechanism and not as a detection advantage and detectors are separated on residual semantic outage, which falls from 73.69 percent to 0.73 percent and on reconstruction fidelity, which rises from 15.97 dB to 21.74 dB. The control traffic share is reduced by 12.62 percent, while the higher absolute signalling and radio energy associated with task preservation are reported explicitly.
8. A five-tier parity study and a zero-shot generalisation study are reported. The parity study supplies multivariate and supervised baselines with the identical features, seeds, labels, protected probe and candidate budget. The generalisation study applies the frozen model and frozen gates, without retraining, to unseen mismatch families, unseen link models and larger codebooks.

The remainder of the paper is organised as follows. Section 2 reviews learned semantic transceivers, digital codebooks, knowledge alignment, sequential monitoring, network efficiency and distributed reasoning. Section 3 defines the endpoint architecture, common knowledge state, semantic representation, physical link and functional mismatch. Section 4 develops the channel-conditioned evidence, protected confirmation, learned beliefs and guarded BDIx decision process. Section 5 presents the implementation, statistical protocol, detector reliability, state-of-the-art comparisons, network metrics, ablations, sensitivity analysis and limitations. Section 6 concludes the study and identifies the required extensions. Appendix A consolidates the symbols used throughout the formulation and evaluation.

2. Literature Review and Background Information

This section reviews learned semantic transceivers, digital semantic codebooks, semantic knowledge alignment, sequential monitoring and distributed agent reasoning together to establish the technical context. The reviewed approaches are separated according to the problem that is solved because representation learning, knowledge alignment, statistical detection and repair control are not interchangeable functions. The broader transition toward native semantic networking and shared semantic knowledge is discussed in [23].

2.1. Learned Semantic Transceivers

Task-oriented semantic transceivers established the foundation for learned transmission. DeepSC was introduced as a transformer-based text semantic communication system in which semantic and channel encoders are trained jointly [1]. Its experiments showed that sentence-level semantic similarity can be preserved more effectively than by conventional separate coding under adverse channel conditions. Deep joint source and channel coding was developed for wireless image transmission and was shown to degrade gracefully as the channel deteriorates [2]. The absence of a rigid source-channel boundary allows the

latent representation to adapt to the channel, although a shared transceiver interpretation is normally assumed.

Robustness at the semantic transceiver level has also received significant attention. Knowledge-enhanced orthogonal frequency division multiplexing was investigated to connect learned semantic features with a digital multicarrier link [24]. A robust semantic text communication system was proposed to address semantic impairments and was reported to improve robustness under semantic noise and channel variation [25]. These methods strengthen semantic recovery, although they do not provide an explicit online diagnosis of receiver codebook divergence.

2.2. Digital Codebooks and Generative Semantic Communication

Digital codebooks are directly relevant to common knowledge because an identical transmitted index must retain an identical meaning at both endpoints. Vector quantised latent learning maps an encoder output to the nearest learned codeword and therefore converts a continuous semantic representation into a discrete index [3]. VQ DeepSC was proposed as a digital image semantic communication system and was reported to provide robustness relative to conventional digital image coding with reconstruction quality comparable to analog semantic transmission in selected conditions [4]. A discrete index is compatible with existing digital links, although the meaning of that index depends on a shared codebook.

Recent codebook-assisted generative systems extend this direction. MOC RVQ combines a multihead octonary codebook with residual vector quantisation and a learned restoration block [26]. The source study reported comparable performance to an analog joint source channel scheme while using approximately one-sixth of its channel bandwidth ratio. A low-rate conditional generative system transmits a codebook-indexed saliency condition together with a semantic text description [27]. The source study reported strong saliency fidelity under an average compression ratio of approximately 0.57 per mille. These results demonstrate the efficiency of shared codebooks, while also increasing the importance of detecting when the codebook meanings have diverged.

2.3. Semantic Knowledge Alignment and Sharing

Semantic alignment mechanisms have been investigated through cooperative learning, partial model adaptation and knowledge exchange. Transceiver cooperative learning was proposed to overcome mismatched background knowledge by periodically exchanging quantised model updates between the endpoints [5,28]. Improved task performance under data mismatch was reported, although communication cost and convergence time depend on repeated collaborative training. Split learning with layer freezing was proposed for resilient multiuser semantic communication [6]. Classification accuracy, reconstruction error and semantic recovery time were used in the source evaluation to show that partial fine-tuning can align heterogeneous neural transceivers.

Direct equalisation and knowledge transfer provide complementary alignment strategies. A semantic channel equaliser based on optimal transport was proposed for language and representation mismatch [7]. A frame-based zero-shot equaliser was later introduced to provide alignment without task-specific retraining [29]. Cooperative semantic knowledge base updating was proposed for multiple semantic pairs through local knowledge selection, global aggregation and pair-specific refinement [8]. The source results indicated larger benefits for initially weak pairs with limited update overhead. Distillation-enabled knowledge alignment was designed for AI agent networks so that compact knowledge can be transferred without complete model exchange [9].

Alignment has also been studied at the network level. A unified hierarchical semantic knowledge base was proposed for multitask communication so that knowledge can be organised horizontally across related tasks and vertically across semantic granularity [30]. Knowledge-sharing-enabled hybrid semantic bit communication was formulated for multi-cell networks [10]. The reported optimisation jointly selected knowledge sharing, semantic extraction and base station association to improve the generalised effective semantic rate under task accuracy and delay constraints. These approaches decide how knowledge should be shared, although an independent protected confirmation stage before repair is not provided.

2.4. Sequential Monitoring and Reliability

The controlled quantitative comparison includes established sequential monitoring methods. CUSUM accumulates positive deviations above a reference slack and is effective when a persistent mean shift is present [11]. Page-Hinkley estimates a running mean and detects a sufficiently large cumulative deviation from its historical minimum [12]. Both procedures are computationally efficient, although their thresholds can be sensitive to nonstationary channel conditions.

Modern anytime-valid alternatives provide stronger statistical guarantees. E detectors provide a nonparametric sequential framework in which false alarm behaviour can be controlled through e values [13]. Conformal test martingales transform online nonconformity information into evidence against exchangeability [14]. Diagnostic runtime monitoring uses class local martingale evidence to identify distributional failures [15]. WATCH extends conformal monitoring through weighted adaptive martingales [16]. Conditional conformal test martingales use contextual conditioning so that distribution shifts can be tested relative to covariate information [17]. These methods supply strong statistical monitoring mechanisms, although a generic alarm does not by itself establish that a semantic repair is safe or necessary.

2.5. Machine Learning for Belief Revision and Risk Estimation

Machine learning can strengthen belief revision when evidence sources interact nonlinearly. Gradient boosting constructs an additive ensemble in which successive weak learners reduce the residual error of the current predictor [31]. Histogram-based gradient boosting extends this principle through efficient discretised split search and supports interactions among heterogeneous continuous and categorical context variables [32]. These properties are relevant when semantic mismatch evidence depends jointly on channel quality, accumulated history, semantic class locality and an active diagnostic response.

Probabilistic belief estimation must nevertheless remain distinct from autonomous action selection. A learned detector can provide strong discrimination, although a direct mapping from an opaque score to a costly repair can weaken safety and auditability. The proposed framework therefore uses machine learning only to update the belief that functional semantic mismatch is present. A protected probe supplies active evidence and the final action remains governed by explicit BDI_x guards. This division preserves the predictive benefit of nonlinear learning while preventing the learner from bypassing channel safety, control budget and explanation requirements.

2.6. Network Efficiency and Green Semantic Communication

Rate, spectral use and energy have increasingly been evaluated alongside task accuracy in semantic communication. Energy-efficient semantic communication with rate splitting was formulated by Yang et al. through the joint control of semantic extraction, common and private transmission, computation frequency and transmit resources [33]. Communication energy and computation energy were considered jointly under latency

and power constraints. The study established that semantic compression can reduce radio traffic, although the computation required to extract and reconstruct semantic information must also be included in the energy balance.

Semantic energy efficiency and adaptive forwarding provide an event-triggered perspective. Tang et al. introduced on-demand semantic forwarding for relay channels and defined semantic energy efficiency as a relationship between recovered semantic utility and transmit energy [34]. Forwarding was admitted only when a semantic similarity check indicated that assistance was needed. Comparable semantic recovery to always-on forwarding was reported with reduced energy use, which demonstrates the value of event-triggered semantic control.

Network-level green semantic communication extends the analysis to infrastructure and computation. Zheng et al. studied aerial-aided edge networks in which communication load is exchanged for computation load and energy efficiency is optimised while proportional service fairness is maintained [35]. Chen et al. derived a generalised effective semantic transmission rate for hybrid semantic and bit communication with knowledge sharing, semantic extraction and small base station association [10]. MOC-RVQ and low-rate codebook generation additionally show that digital semantic representations can reduce channel bandwidth when codebooks and generative priors are available [26,27].

A measurement gap remains because the available studies use different tasks, semantic utility functions, channel models, bandwidths and energy boundaries. Their numerical values therefore cannot be transferred into one table as if they were measured under identical conditions. CoSMA DAI therefore reports two complementary comparisons. The literature table records the metric and outcome claimed by each original source, while the controlled evaluation computes task-weighted data rate, bandwidth-normalised semantic spectral efficiency, control-adjusted semantic resource efficiency, radio energy and control traffic share for all implemented monitors on identical streams and under one disclosed physical accounting model.

2.7. Distributed Artificial Intelligence and Research Gap

The proposed control loop is grounded in distributed reasoning. The Belief Desire Intention model separates informational state, objectives and committed actions [18]. Multiagent learning extends this structure to distributed decision settings [19]. A distributed artificial intelligence framework for communication mode selection was developed in [20]. A broader BDIX communication architecture and an intelligent plan library were presented in [21,22]. These studies support auditable communication control, but semantic codebook divergence and protected repair confirmation were not addressed.

Despite these advances, an integrated gap remains. Representation methods transmit learned semantics. Alignment methods update or equalise semantic knowledge. Sequential methods detect statistical change. Agent frameworks structure distributed decisions. A unified protocol is still required to determine whether a change is semantic rather than channel-induced, to request independent evidence, to reject unsafe repair under poor channel conditions and to record the reason for the selected action. Table 1 compares the principal approaches and summarises the results reported by their original sources without transferring numerical values across incompatible datasets.

Table 1. Comparison of representative state-of-the-art semantic communication, alignment and monitoring approaches.

Approach	Core Mechanism	Result Described in the Source	Explicit Mismatch Handling	Online Repair Decision	Main Gap Relative to CoSMA DAI
DeepSC [1]	Joint text semantic and channel coding	Higher sentence semantic similarity than conventional separate coding under adverse channels	No	No	Stable endpoint interpretation is assumed
DeepJSCC [2]	End-to-end image source channel coding	Graceful reconstruction degradation and strong low channel quality performance	No	No	Codebook divergence is not diagnosed
VQ DeepSC [4]	Digital vector-quantised semantic indices	Robust digital image transmission with quality comparable to analog semantic coding in selected settings	Indirectly	No	Shared codeword meaning is assumed
Robust semantic text [25]	Robust semantic encoding against semantic impairment	Improved semantic text recovery under semantic noise and channel variation	Partly	No	No protected confirmation before model repair
MOC RVQ [26]	Multilevel codebook with residual quantisation	Comparable performance to analog JSCC with approximately one sixth channel bandwidth ratio	No	No	Codebook synchronisation is assumed
Low rate codebook generation [27]	Codebook-indexed saliency and semantic text prompts	Strong saliency fidelity at an average compression ratio near 0.57 per mille	No	No	No online drift monitor is provided
Energy-efficient RSMA semantic communication [33]	Joint semantic extraction, rate splitting and resource allocation	Reduced joint communication and computation energy under latency and power constraints	Knowledge is shared	Optimization-based	A semantic drift detector and protected repair confirmation are not provided
On-demand semantic forwarding [34]	Relay assistance triggered by semantic similarity checks	Comparable recovery to always-on forwarding with lower energy consumption	Indirectly	Forwarding decision	Codebook mismatch diagnosis and endpoint repair are not addressed
Aerial-aided green semantic communication [35]	Joint semantic computation, radio allocation and service fairness	Improved energy efficiency while maintaining proportional service revenue	No	Optimisation based	The model targets resource efficiency rather than online semantic alignment
Cooperative learning [5,28]	Periodic collaborative transceiver updates	Improved task performance under endpoint data mismatch	Yes	Scheduled	Repeated training and update communication are required
Split learning alignment [6]	Partial layer fine tuning with layer freezing	Improved accuracy, reconstruction error and semantic recovery time under heterogeneous transceivers	Yes	Trigger is method specific	No independent semantic probe is used

Table 1. Cont.

Approach	Core Mechanism	Result Described in the Source	Explicit Mismatch Handling	Online Repair Decision	Main Gap Relative to CoSMA DAI
Semantic channel equaliser [7]	Optimal transport-based representation equalisation	Improved recovery under language or representation mismatch	Yes	Equalisation is direct	Channel confounding is not separately confirmed
Cooperative SKB update [8]	Local knowledge selection and global SKB refinement	Larger gains for weak pairs with limited update overhead	Yes	Update policy	Repair is not preceded by protected confirmation
Knowledge sharing hybrid link [10]	Joint knowledge sharing, semantic extraction and association	Improved effective semantic rate under accuracy and delay constraints	Yes	Optimisation-based	Online statistical drift diagnosis is not central
WATCH and CTM methods [14,16,17]	Anytime valid martingale monitoring	Controlled false alarm monitoring and adaptive change evidence	Statistical shift	Alarm only	Semantic repair logic and message cost are absent
Gradient boosted belief estimation [31,32]	Nonlinear supervised probability estimation from heterogeneous evidence	Strong interaction modelling with efficient tree ensembles	Depends on labels	No direct repair logic	Safety guards and active semantic confirmation are not inherent
CoSMA DAI	Conditioned evidence, learned belief revision, protected probe and BDIX guarded repair	Controlled results are reported in Section 5 with accuracy, precision, recall, F1, MCC, delay, reliability and overhead	Yes	Yes	Larger tasks and hardware evaluation remain future work

3. System Description and Problem Formulation

The system formulation combines the distributed endpoint architecture, learned semantic codebook, mobile physical link and functional semantic mismatch model. The variables required by the proposed method are summarised in the notation table.

3.1. Distributed Endpoint Architecture and Protocol

The endpoint architecture comprises transmitter and receiver agents, a mobile network environment and an explicit control-message protocol. A transmitter BDIX agent and a receiver BDIX agent are placed at the two endpoints. The acknowledged common state is represented by Equation (1).

$$\mathcal{K}_t^{\text{ack}} = \langle \mathcal{C}_t^{\text{ack}}, \mathcal{T}, v_t, a_t, \chi_t \rangle, \quad (1)$$

where $\mathcal{C}_t^{\text{ack}}$ is the last acknowledged codebook, $\mathcal{T}$ is the task, v_t is the version, a_t is the acknowledgement state and χ_t is the alignment confidence. Version equality is not treated as proof of functional equivalence.

Figure 1 presents the complete mobile communication architecture. The updated infographic labels the network elements and the arrow flows directly, including payload or semantic flow, uplink index or symbol flow, downlink probe and control flow and control or feedback flow. A semantic codeword index, a compact fingerprint and an authenticated task belief are delivered on the payload path. The receiver updates global evidence, semantic class local evidence and a machine-learned mismatch belief. A protected probe request

is issued only after a passive candidate gate has been reached. A known anchor is then returned by the transmitter and its response is incorporated into the belief update. In the principal first-repair policy, a repair request, the resynchronisation payload and an acknowledgement are exchanged only when the learned belief and channel-safety guard are satisfied. The dedicated recurrent stress policy is distinguished in the plan description. The complete cost of every exchange is included in Table 2.

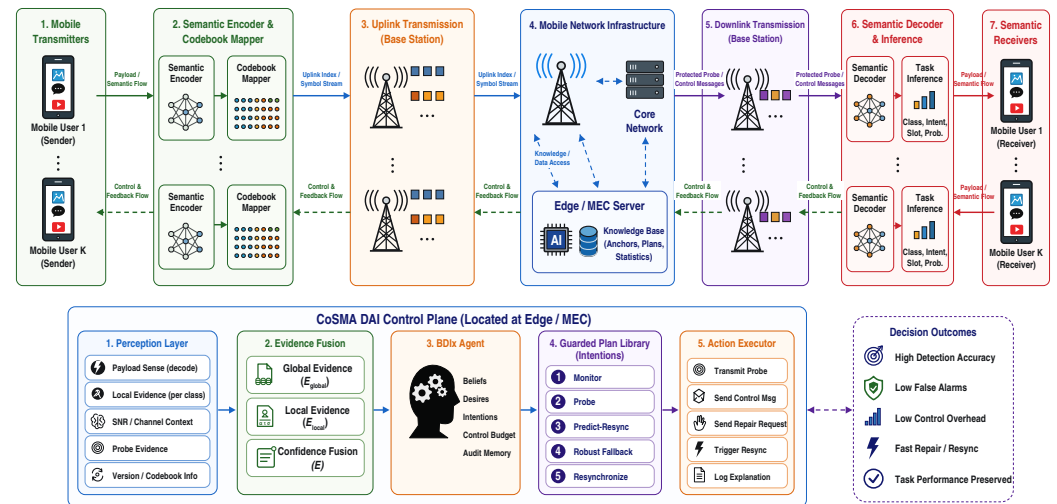


Figure 1. CoSMA DAI system architecture in a mobile communication environment.

Table 2. Control messages included in the protocol accounting.

Exchange	Component Bits	Total Bits	Function
Task belief header	4×3	12	Repetition coded source task belief on each semantic packet
Task belief rescue	$(4 + 4) \times 5$	40	Protected task belief and cyclic redundancy check after payload rejection
Protected probe	$64 + 384 + 64$	512	Probe request, protected semantic anchor and acknowledgement
Full resynchronisation	$64 + 8192 + 64$	8320	Repair request, complete codebook update and acknowledgement

3.2. Learned Semantic Codebook and Task Model

The semantic representation, codebook mapping and downstream task model are defined as follows. A source observation x_t is transformed into the latent vector defined in Equation (2).

$$z_t = f_\theta(x_t) \in \mathbb{R}^{d_z}. \quad (2)$$

A learned codebook $\mathcal{C}^{\text{tx}} = \{c_k^{\text{tx}}\}_{k=1}^{N_c}$ is formed by clustering the training latent vectors. The transmitted codeword is selected through Equation (3).

$$k_t = \arg \min_{k \in \{1, \dots, N_c\}} \|z_t - c_k^{\text{tx}}\|_2^2. \quad (3)$$

A task classifier g_ψ is applied to the receiver codeword. The aligned task output is $\hat{y}_t = g_\psi(c_{k_t}^{\text{tx}})$. A transmitter task posterior $\pi_t^{\text{task}} = h_\omega(z_t)$ is also formed from the source latent representation. Its highest probability class $\hat{y}_t^B = \arg \max_j \pi_{t,j}^{\text{task}}$ is maintained as compact common knowledge. This task belief is used only to preserve the downstream decision, while complete semantic alignment is being confirmed and repaired. A compact finger-

print is produced by a fixed orthonormal projection $R \in \mathbb{R}^{d_f \times d_z}$ and row normalisation. The fingerprint associated with codeword k is denoted by q_k .

3.3. Physical Link Model

A mobile wireless link model separates physical corruption from semantic divergence. The received signal power is modelled through distance-dependent path loss, correlated shadowing and Rician fading. The instantaneous signal-to-interference-plus-noise ratio is defined in Equation (4).

$$\Gamma_t = \frac{P_{\text{tx}} |h_t|^2 L(d_t)^{-1}}{I_t + N_0 B}, \quad \gamma_t = 10 \log_{10} \Gamma_t, \quad (4)$$

where P_{tx} is the transmit power, h_t is a Rician coefficient with factor $K = 4$, $L(d_t)$ is path loss, I_t is autoregressive interference power, N_0 is the noise spectral density and B is bandwidth. The semantic index is represented by seven bits because $N_c = 128$. The binary phase shift keying approximation and repetition residual are defined in Equation (5).

$$p_b(\Gamma_t) = Q(\sqrt{2\Gamma_t}), \quad p_b^{(3)} = 3p_b^2 - 2p_b^3, \quad (5)$$

where $p_b^{(3)}$ is the residual bit error probability after threefold repetition. An eight-bit cyclic redundancy check is applied. Detected packet errors are rejected and rare undetected errors remain possible after the cyclic redundancy check. Therefore, channel corruption can affect the semantic index, but a separate codebook mismatch can still occur when the index is delivered correctly.

3.4. Functional Semantic Mismatch

Functional semantic mismatch and the associated control objective are defined using the receiver codebook state. The receiver codebook is denoted by $\mathcal{C}_t^{\text{rx}}$. A functional semantic outage is declared through Equation (6).

$$\mathcal{O}_t^{\text{sem}} = \left\{ g_\psi(c_{k_t,t}^{\text{rx}}) \neq g_\psi(c_{k_t}^{\text{tx}}) \right\}, \quad (6)$$

conditional on syntactic acceptance. Abrupt divergence is created by mapping each codeword to the nearest codeword assigned to the next task class. Gradual divergence is formed by interpolation between the aligned and mapped codewords. Sparse divergence is restricted to codewords assigned to classes 3, 5 and 8. Under gradual divergence, the delay origin is the first slot at which the interpolation strength reaches 0.10. This definition prevents a delay from being reported before a measurable codebook displacement exists.

The control objective is to detect and independently confirm mismatch, avoid repair when the channel is unsafe and minimise task loss and control traffic. No general optimisation expression is introduced because the executable method is a fixed guarded policy rather than a solved numerical programme. The complete symbols and notation are consolidated in Appendix A.

4. Methodology

This section provides the complete CoSMA DAI methodology, which combines passive evidence, protected confirmation, learned belief revision, task preservation and guarded repair. Channel-conditioned semantic evidence is first constructed and global and semantic class local memories are then updated. A passive candidate gate requests a protected probe. The payload, temporal, channel and probe observations are subsequently converted into a learned semantic mismatch belief. In parallel, a transmitter task posterior is retained as

compact common knowledge so that the downstream task can be protected while receiver semantics remain uncertain. A guarded BDix intention is finally selected without allowing either learned belief to bypass the channel safety and control constraints. Experimental configuration and computational details are reserved for Section 5.

The design follows a separation of evidence, belief and action. Evidence variables record what has been observed from payload packets and protected probes. Belief variables express the estimated semantic state and the available task-preserving information. Intentions specify the admissible communication action after safety, recurrence and budget guards have been evaluated. This separation prevents a transient score excursion from being interpreted simultaneously as semantic mismatch, confirmed diagnosis and permission to transfer a complete codebook. It also makes every transition auditable because the evidence that opened the candidate stage, the probe outcome that revised the belief and the guards that admitted or rejected repair can be recorded independently.

Online operation is organised as a finite state interaction between normal continuation, candidate evaluation, protected confirmation, guarded fallback and semantic resynchronisation. Only accepted payload packets contribute new semantic evidence; global and local memories continue to decay on every physical slot. A candidate can request active confirmation but cannot alter the acknowledged common codebook state. The common state is changed only after a repair message, a codebook transfer and an acknowledgement have completed successfully. The task preservation belief operates across these states as a temporary application safeguard rather than as a substitute semantic representation.

4.1. Semantic Consistency Score and Context Conditioning

The control loop is driven initially by payload evidence. The payload fingerprint is affected by the physical channel and is compared with the fingerprint associated with the receiver codeword. The semantic consistency score is defined in Equation (7).

$$s_t = \frac{\|q_{k_t} + n_t - q_{\tilde{k}_t}^{\text{rx}}\|_2}{\sqrt{d_f}}, \quad (7)$$

where n_t denotes fingerprint perturbation and $\tilde{k}_t$ denotes the accepted decoded index. A rejected packet contributes zero contextual excess, while the evidence memories continue to decay and the candidate gate remains evaluated on the physical-slot clock.

Channel conditioning is implemented through separate reference distributions for every signal-to-interference-plus-noise-ratio stratum and transmitted codeword. For an accepted payload packet, the clipped positive contextual excess is defined in Equation (8).

$$z_t^{\text{ctx}} = \min \left\{ z_{\max}, \left[\frac{s_t - m_{b_t, k_t}}{\max(r_{b_t, k_t}, \epsilon_r)} \right]_+ \right\}, \quad (8)$$

where b_t denotes the active channel stratum, $m_{b, k}$ denotes the reference median, $r_{b, k}$ denotes the robust scale and ϵ_r denotes a positive scale floor. Here $[u]_+ = \max\{u, 0\}$ and $z_{\max}$ is the upper bound applied to both payload and protected-probe evidence. A rejected payload contributes $z_t^{\text{ctx}} = 0$. The scale is obtained from the larger normalised value produced by the median absolute deviation and the interquartile range. This conditioning suppresses score movements that are typical for the current channel and preserves positive deviations that are atypical for both the channel and the transmitted semantic identity.

Reference statistics are indexed jointly by the transmitted codeword and the channel stratum because different semantic fingerprints can have different natural reconstruction distances and different channel conditions can produce different perturbation scales. Median-based location and robust scale estimates are used so that a small number of

development outliers cannot inflate the normal operating region. The positive part operator removes evidence in the direction of improved agreement and retains only excess inconsistency. A scale floor prevents unstable normalisation when a reference cell has very low dispersion. Consequently, the detector input represents contextual semantic surprise rather than raw Euclidean distance.

Packet acceptance is treated as a prerequisite for evidence accumulation. A cyclic redundancy check failure indicates that the physical layer has not delivered a trustworthy semantic index, so the corresponding observation is handled by the communication protocol but is not interpreted as evidence of codebook divergence. This distinction avoids a common confounding path in which poor channel delivery would be counted simultaneously as physical corruption and semantic mismatch.

4.2. Global and Semantic Class Local Evidence

Two complementary evidence memories are maintained. A global accumulator is used to capture persistent drift that affects many semantic classes. A class local accumulator is used to preserve repeated evidence associated with a specific semantic meaning. Their updates are given by Equations (9) and (10).

$$G_t = \rho G_{t-1} + [z_t^{\text{ctx}} - \kappa_g]_+, \quad (9)$$

$$L_{j,t} = \rho L_{j,t-1} + \mathbb{I}\{\hat{y}_t = j\} [z_t^{\text{ctx}} - \kappa_l]_+. \quad (10)$$

Every local accumulator is decayed at each slot so that evidence associated with an inactive semantic class does not remain indefinitely. The active evidence is defined in Equation (11).

$$E_t = (1 - w_l)G_t + w_l L_{\hat{y}_t,t}, \quad (11)$$

where w_l controls the balance between stream-wide persistence and semantic class locality. The two memories are retained because a broad shift and a sparse class-dependent shift produce different temporal patterns.

The passive candidate gate is defined in Equation (12). A candidate is admitted when fused evidence remains at or above its gate for c_E consecutive physical slots or when semantic class local evidence reaches the dedicated local gate.

$$C_t = \mathbb{I}\left\{ \bigcap_{u=t-c_E+1}^t \{E_u \geq h_E\} \right\} \vee \mathbb{I}\{L_{\hat{y}_t,t} \geq h_L\}. \quad (12)$$

The local route prevents sparse divergence from being diluted by classes that remain aligned, while the persistent fused route rejects isolated broad fluctuations.

The accumulator design produces different responses for the three evaluated divergence patterns. Abrupt mismatch raises both global and active local evidence quickly because many consecutive accepted packets become atypical. Gradual mismatch is integrated over time until persistent contextual excess exceeds the candidate requirement. Sparse mismatch can remain weak in the global stream because most semantic classes are still aligned, while the local accumulator associated with an affected class retains repeated evidence whenever that class reappears. Exponential decay limits the memory horizon and allows the state to return toward normal when atypical observations cease.

Persistence is evaluated over consecutive physical slots. A rejected packet supplies no new contextual excess, but the stored evidence is decayed and the persistence gate is evaluated. A candidate can therefore remain active after a packet rejection when its retained evidence is still sufficiently large; protected confirmation remains subject to the channel-safety guard. The two candidate routes also separate sensitivity from safety. The local gate

provides sensitivity to concentrated class-dependent mismatch, while the persistent fused gate requires repeated evidence before broad repair activity can be considered.

4.3. Protected Probe Confirmation

Active confirmation is initiated after a passive candidate has satisfied the channel-safety condition. When $C_t = 1$, a protected probe request is sent if the current signal-to-interference-plus-noise ratio satisfies the safety guard. The transmitter returns a known semantic anchor associated with the same acknowledged codeword. The received probe score is normalised through a reference table that is independent from the payload calibration. The protected evidence is defined in Equation (13).

$$P_t = \min \left\{ z_{\max}, \left[\frac{s_t^{\text{probe}} - m_{b_t^{\text{probe}}, k_t}^{\text{probe}}}{\max(r_{b_t^{\text{probe}}, k_t}^{\text{probe}}, \epsilon_r)} \right]_+ \right\}. \quad (13)$$

The index b_t^{probe} identifies the channel stratum of the protected-probe response, which can differ from the payload stratum b_t . A payload fluctuation can therefore open the diagnostic stage, although it cannot directly commit the expensive repair. The protected response becomes available only after the probe has been requested. It is then supplied to the learned belief updater together with the passive and contextual evidence.

The probe exchange provides an intervention that is operationally different from passively observing another application packet. A request identifies the acknowledged semantic codeword and asks the transmitter to return a known anchor under protected coding. Because the expected semantic content is known in advance, disagreement in the returned anchor is more diagnostic than a payload observation whose source variation is uncontrolled. Probe reference statistics are learned separately from payload statistics so that the stronger coding and the restricted anchor distribution do not create an invalid calibration shortcut.

Channel safety is checked before the request because active confirmation under a severely degraded link would add cost without producing independent evidence. When the safety condition is not met, robust fallback preserves the current acknowledged state and postpones repair. When the probe is completed, its score contributes to belief revision but still does not alter the codebook directly. The staged design therefore converts passive suspicion into active evidence without allowing the confirmation message itself to become an automatic resynchronisation trigger.

4.4. Machine-Learned Semantic Mismatch Belief

The learned mismatch belief estimates whether the endpoint semantic meanings have diverged. The causal feature vector contains the current conditioned and raw scores, protected probe evidence, signal-to-interference-plus-noise ratio, packet acceptance, global evidence, active class local evidence, fused evidence, their ratios and differences, first differences and short rolling summaries. It is represented compactly in Equation (14).

$$\mathbf{x}_t^B = \Phi(z_{1:t}^{\text{ctx}}, s_{1:t}, P_t, \gamma_t, A_t^{\text{pay}}, G_{1:t}, L_{g_{1:t}}, E_{1:t}). \quad (14)$$

Only present and past observations are contained in $\mathbf{x}_t^B$. The indicator A_t^{pay} equals one when the payload is accepted and zero otherwise; it is distinct from the common-state acknowledgement a_t . In online control, the active probe component is consumed only after a channel-safe candidate has requested the probe and the exchange has been completed. Future semantic labels, future channel states and held-out outcomes are not included.

The learned mismatch score is defined in Equation (15).

$$B_t = f_\theta(\mathbf{x}_t^B) \in [0, 1], \quad (15)$$

where f_θ denotes a supervised nonlinear score estimator learned only from development streams. The development target is $\ell_t^{\text{div}} = \mathbb{I}\{\alpha_t \geq 0.10\}$, where α_t is the imposed interpolation strength for the codeword processed at slot t and is zero for unaffected codewords in sparse-divergence streams. Matched-null samples have $\ell_t^{\text{div}} = 0$. This target identifies imposed codebook divergence and is distinct from the task-output disagreement event $\mathcal{O}_t^{\text{sem}}$. The target is used for supervised development and offline evaluation; it is not supplied to the online controller. The output is used as a decision score at a fixed operating point and is not relied upon as a calibrated posterior. Its calibration against the imposed-divergence target is measured in Section 5.10: the expected calibration error is 0.00037 and the Brier skill score is 0.9980, but the maximum calibration error over sparsely populated intermediate bins reaches 0.704. The estimator models interactions that cannot be represented by one fixed probe threshold, such as a large probe deviation under a poor channel, moderate probe evidence after persistent class local accumulation or a sudden score increase that is inconsistent with recent channel history. The score is treated as one BDIx belief rather than as an autonomous action command.

The feature construction can be interpreted through five groups. Instantaneous semantic features describe the raw and conditioned payload score. Memory features describe global, local and fused evidence. Temporal features describe first differences and short rolling summaries. Channel features describe signal-to-interference-plus-noise ratio and packet acceptance. Active confirmation features describe whether a probe has been completed and how strongly its response departs from the independent probe reference. Combining these groups allows the estimator to distinguish, for example, a large raw score caused by a poor channel from a sustained class local inconsistency that remains after protected confirmation.

Model training is separated from gate selection. Candidate estimators are selected on development groups and the frozen estimator produces causal slot-level mismatch scores. The belief threshold is then selected with the guarded evidence parameters on a disjoint development group. Held-out drift streams and both null reliability campaigns remain excluded from these decisions. This sequence prevents the evaluation streams from influencing either the nonlinear discriminator or the operating point at which a belief becomes actionable.

A learned task-preservation belief is defined in parallel. The source posterior and its compact decision are given in Equation (16).

$$\pi_t^{\text{task}} = h_\omega(z_t), \quad \hat{y}_t^B = \arg \max_j \pi_{t,j}^{\text{task}}. \quad (16)$$

The posterior is produced from the transmitted semantic representation and does not use the hidden test label. The compact decision is stored in the acknowledged common state and is made available to the receiver as auxiliary task evidence. A divergent receiver codebook can therefore alter the reconstructed latent meaning without immediately forcing the downstream task to follow that altered mapping.

The task-output fusion rule is given in Equation (17).

$$\hat{y}_t^{\text{out}} = \begin{cases} \hat{y}_t^B, & A_t^B = 1, \\ g_\psi(c_{k_t,t}^{\text{rx}}), & A_t^B = 0, \end{cases} \quad (17)$$

where A_t^B indicates that the compact task belief has been authenticated and accepted. The complete semantic representation is still repaired through codebook resynchronisation. The task belief only provides a temporary task-oriented fallback and its communication cost is charged explicitly in the evaluation.

The task preservation path is deliberately narrower than full semantic repair. Only the compact decision required by the current downstream task is transmitted, while the receiver codebook and latent representation remain unchanged until resynchronisation is confirmed. Authentication is required because an unauthenticated auxiliary decision could corrupt the application output even when the semantic codebooks are aligned. The ordinary task belief header and any rescue token required after packet rejection are counted as semantic side information because they directly convey useful task information and remain included in total radio resources and energy.

A distinction is therefore maintained between representation recovery and task continuity. Codebook resynchronisation restores a common semantic representation for future messages. The task belief maintains the current application decision during the interval in which that representation cannot yet be trusted. The ablation in Section 5.7 measures this distinction directly by removing the task belief while preserving the mismatch detector and repair logic.

4.5. BDIx Beliefs, Desires and Guarded Intentions

The deliberation structure combines the receiver belief state, ordered desires and guarded intentions. The receiver beliefs contain the acknowledged codebook version, channel state, global evidence, semantic class local evidence, protected evidence, learned mismatch score, authenticated task belief, recurrence state and remaining control budget. The desires are ordered as semantic correctness, false repair avoidance, channel safety, task preservation, low control cost and explanation completeness. The principal first-repair evaluation uses the guarded intention rule in Equation (18), where $\mathcal{I}_t$ is the selected intention and $\gamma_{\min} = 2$ dB is the channel-safety threshold. At a channel-safe candidate, a protected probe is obtained subject to a five-slot probe cooldown; the probe outcome is retained when the belief remains below its gate and full resynchronisation is selected when the gate is reached.

$$\mathcal{I}_t = \begin{cases} \mathcal{I}_{\text{cont}}, & C_t = 0, \\ \mathcal{I}_{\text{safe}}, & C_t = 1, \gamma_t < \gamma_{\min}, \\ \mathcal{I}_{\text{probe}}, & C_t = 1, \gamma_t \geq \gamma_{\min}, B_t < \theta_B, \\ \mathcal{I}_{\text{sync}}, & C_t = 1, \gamma_t \geq \gamma_{\min}, B_t \geq \theta_B. \end{cases} \quad (18)$$

Continuation preserves the current semantic state and robust fallback postpones repair under an unsafe channel. The dedicated recurrent stress experiment uses a separate history-dependent policy. Its candidate condition is $E_t > h_E$, without the principal persistence or local-route gate. After an earlier repair and a gap of more than 30 slots, a channel-safe candidate activates predictive resynchronisation from retained evidence without a fresh protected probe or a new test of $B_t \geq \theta_B$. Otherwise, the candidate follows the protected-probe and learned-belief path. This stress policy measures history-based plan activation and does not establish independent protected confirmation for every recurrent repair.

Learning and deliberation are deliberately separated. The machine learning component estimates one belief variable. The plan library maps the complete belief state to an admissible action through fixed guards. An equivalent threshold cascade is therefore expected to reproduce the plan decision after B_t has been supplied. The plan representation

is retained because an explicit intention, guard state, message cost and explanation can be associated with every decision.

Belief revision and intention selection are executed in a fixed order. Newly accepted evidence updates the semantic and channel beliefs. The desire ordering then prioritises semantic correctness and false repair avoidance before communication economy. Candidate plans whose guards are false are removed from consideration. The remaining intention determines whether normal communication continues, confirmation is requested, repair is postponed, recurrent mismatch is handled predictively or full resynchronisation is committed. This ordering ensures that an increase in the mismatch score cannot suppress an unsafe channel condition or exceed the remaining control budget.

An explanation record can contain the acknowledged codebook version, candidate route, global and local evidence, protected probe result, learned mismatch score, task belief status, channel guard, recurrence flag, selected intention and charged messages. Such records do not make the learned estimator intrinsically interpretable, although they make the protocol decision traceable. The explanation identifies which evidence and guards were used when an action was taken and separates model output from the final guarded decision.

4.6. End to End Decision Procedure

The complete decision sequence proceeds as follows. First, an accepted semantic index and fingerprint are used to obtain z_t^{ctx} and the authenticated source task belief is recorded when it is available. Second, G_t , all $L_{j,t}$ and E_t are updated. Third, C_t is evaluated through persistent fused evidence and the semantic class local gate. Fourth, robust fallback is selected when a candidate is raised under an unsafe channel. Fifth, a protected probe is requested when the channel is admissible. Sixth, the probe response and all causal passive features are converted into B_t . Seventh, the task output is preserved through Equation (17) while full semantic repair remains pending. Eighth, full resynchronisation is committed only when $B_t \geq \theta_B$ and the remaining BDIx guards are satisfied. Finally, the acknowledged common state is updated and the evidence memories are reset after a completed repair.

Detection and control remain distinct stages. The passive gate identifies a candidate semantic inconsistency. The protected probe obtains independent evidence. The learned updater revises the mismatch belief. The BDIx guards determine whether an action is admissible. The repair message changes the shared semantic state. These stages are separated so that a high machine learning score is not treated as equivalent to a confirmed and safe communication action.

During ordinary continuation, the online state consists of one global accumulator, one local accumulator for each semantic class, a short causal history buffer, the latest task belief and the BDIx protocol state. Per slot processing requires score normalisation, accumulator updates and one belief inference after the required observations are available. Protected probing and codebook transfer are event-driven rather than periodic. The dominant routine cost is therefore the fixed-size belief inference, while the large resynchronisation payload is incurred only after confirmation and guard satisfaction.

State reset after repair is essential for avoiding repeated alarms from evidence accumulated under the obsolete codebook. The acknowledged codebook version is advanced only after the receiver confirms completion. Global, local and probe evidence are then cleared and recurrence history is retained separately so that a later return of mismatch can activate predictive resynchronisation. The resulting procedure preserves causal ordering between observation, confirmation, decision, communication and common state update.

5. Evaluation Results and Discussion

This section provides the implementation, experimental and quantitative details, which are consolidated below. The semantic benchmark, channel generation, machine learning belief estimator, development groups, held-out streams, metrics, state-of-the-art monitor comparison, ablations, stress scenarios and limitations are reported in one section so that methodological concepts remain separated from software and configuration choices.

5.1. Experimental Implementation and Semantic Benchmark

The simulation is implemented in Python with scikit-learn for the learned models, NumPy and SciPy for numerical and statistical routines, Matplotlib for plotting and Pillow for image handling. The archived belief estimator records scikit-learn 1.8.0 [36] and the original plot metadata records Matplotlib 3.10.8. The supplied requirements specify NumPy 1.26 or later, SciPy 1.11 or later and Pillow 10.0 or later. PyTorch and torchvision are used for the pretrained MobileNetV3-small CIFAR features, while JAX and Optax are used for the CPU convolutional VQ-VAE. Exact historical versions of Python, NumPy, SciPy, Pillow, PyTorch, torchvision, JAX and Optax were not retained in the archived experiment records.

The subsequent frozen-model verification was performed with Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0, Matplotlib 3.10.8 and Pillow 12.3.0. The principal delays and task accuracy were reproduced, with agreement between the independent cascade and the controller on all 540 checked streams. CIFAR feature extraction and VQ-VAE training were not repeated in this verification.

The reported results are produced by a complete implementation based on the handwritten-digits data distributed with scikit-learn, which are used as a compact semantic source [36,37]. Pixel vectors are standardised and projected to $d_z = 24$ dimensions through principal component analysis. A 128-entry semantic codebook is learned through mini-batch k-means and a logistic task classifier is learned in the latent space. The held-out task accuracy after codebook quantisation is 0.919. The same classifier produces the transmitter task posterior without access to held-out labels. Its top class is represented by four bits and is piggybacked on the protected semantic packet. When the payload cyclic redundancy check rejects the packet, a fivefold repeated rescue token with a four-bit cyclic redundancy check is returned after the negative acknowledgement. The ordinary task belief header and every attempted rescue token are included in the charged overhead.

The benchmark is used as a controlled semantic communication task rather than as evidence of deployment readiness for large media models. It provides a genuine learned representation, a discrete codebook, multiple semantic classes and an independently measurable downstream task. These properties allow codeword meaning to be changed at the receiver while the physical packet format remains valid. The resulting failure is therefore semantic and functional rather than a synthetic scalar change applied directly to a detector input. The quantised task ceiling also establishes the maximum performance available from the frozen representation before channel and alignment effects are introduced.

The implementation uses $z_{\max} = 30$ and $\epsilon_r = 10^{-4}$ for payload and protected-probe score normalisation. The protected-probe SINR is modelled as $\gamma_t^{\text{probe}} = \min\{22, \max\{-4, \gamma_t + 3\}\}$, with all quantities expressed in dB. Its stratum b_t^{probe} is selected from the same six intervals as the payload: $[-8, -2)$, $[-2, 2)$, $[2, 6)$, $[6, 10)$, $[10, 14)$ and $[14, 22]$ dB.

Each evaluation stream contains 500 slots and follows a controlled divergence process that begins at slot 210. Abrupt divergence is applied immediately. Gradual divergence reaches its evaluation onset when the interpolation strength reaches 0.10. Sparse divergence affects codewords associated with semantic classes 3, 5 and 8. The physical link contains Rician fading with $K = 4$, distance-dependent path loss, correlated shadowing, autore-

gressive interference, binary phase shift keying, threefold repetition and an eight-bit cyclic redundancy check. A matched null stream uses the same channel process without codebook divergence. A degrading ramp stream is retained as an additional channel stress control.

The three harmful scenarios test different failure geometries. Abrupt divergence measures response after a sudden global inconsistency. Gradual divergence tests whether weak but persistent evidence can be integrated without reacting to ordinary fading. Sparse divergence tests whether class local memory can identify affected meanings when most codewords remain aligned. The matched null streams estimate false repair under the nominal channel distribution, while the degrading ramp control stream examine whether a sustained physical deterioration is incorrectly reclassified as semantic mismatch. Identical channel generators and seed-based stream construction are used across methods so that differences arise from the monitoring and control logic rather than from different radio realisations.

Histogram gradient boosting and extremely randomised trees are compared as candidate belief estimators because both can represent nonlinear interactions without requiring future sequence information. Histogram gradient boosting is selected from the development data and is configured with 250 boosting iterations, a learning rate of 0.06, at most 31 leaves, an L_2 regularisation value of 1.5 and balanced class weights. The selected estimator uses 20 causal features. Its development validation area under the receiver operating characteristic curve is 0.99999.

Data leakage is prevented through a grouped development split. Initial model fitting uses 40 drift seeds and 100 matched null seeds. Model selection and belief gate selection use a separate group of 20 drift seeds and 100 matched null seeds. The selected estimator is then refitted on all 60 drift development seeds and 200 null development seeds, which produce 190,000 slot-level training rows. The held-out test seeds, the 300 matched reliability seeds and the 300 degrading channel seeds are excluded from every model and gate selection operation. The selected belief gate is 0.9990, the persistent evidence gate is 7.50, the semantic class local gate is 15.00 and the persistent evidence requirement is 3 consecutive physical slots.

The sequential comparators are configured only on development data and are then frozen for the held-out evaluation. They are compact implementations that follow the cited constructions but omit several elements of the published algorithms: WATCH is realised as an exponentially weighted moving average of the conditioned score with a gate, the Diagnostic E Process as a class-local logarithmic e-process and the two conformal test martingales as power betting on a fixed reference law without randomised ranks, covariate weighting or online betting adaptation. They are therefore referred to as CUSUM-type, Page Hinkley, WATCH-type and CTM-type monitors when the distinction matters. Results of the full paper-defined algorithms are reported separately in the controlled monitor comparison subsection. Their configuration selection is performed on 10 drift development seeds and 30 matched null streams. Every comparator receives 144 disclosed candidates. The learned CoSMA DAI belief layer uses two model candidates followed by 144 guarded gate configurations on the separate development groups described above. Every evaluated method therefore receives the same number of final configuration candidates. This equality concerns candidate counts, not the amount of development data: the principal comparators use 10 drift and 30 matched null development streams, whereas the learned belief uses 60 drift and 200 matched null development streams. The equal-development-budget comparison is reported in Section 5.10. Every configuration is frozen before 30 held-out seeds, 300 matched null streams and 300 degrading channel streams are evaluated. The same conditioned payload score is supplied to all methods in the principal comparison. Table 3 lists the complete development search spaces and confirms the equal final candidate counts.

Table 3. Development search spaces for the compared methods.

Method	Candidate Construction	Count
CoSMA DAI	Two belief estimators; $h_E \in \{7.5, 9.0, 11.0, 13.0\}$, $h_L \in \{12, 15, 18\}$, $\theta_B \in \{0.995, 0.999, 0.9995\}$ and $c_E \in \{1, 2, 3, 4\}$; $\rho = 0.99$ and $w_l = 0.80$	144
CUSUM	12 slack values from 0.10 to 1.20 and 12 gates from 3 to 20	144
Page Hinkley	12 delta values from 0.05 to 0.80 and 12 gates from 3 to 20	144
WATCH	12 decay values from 0.80 to 0.995 and 12 gates from 0.8 to 8	144
Diagnostic E Process	12 betting gains from 0.03 to 0.30 and 12 log evidence gates from 2 to 16	144
Conditional CTM	12 betting exponents from 0.10 to 0.80 and 12 log evidence gates from 2 to 16	144
Reference CTM	12 betting exponents from 0.10 to 0.80 and 12 log evidence gates from 2 to 16	144

Fairness is enforced at both the data and input levels. Development seeds are not reused as held-out test or reliability seeds. Every monitor receives the same accepted packet sequence and the same channel-conditioned scalar score, so no baseline is disadvantaged by receiving an unnormalised observation. Candidate counts are equalised at the final tuning stage and the selected configuration is frozen before any reported stream is processed. The complete search ranges, selected configuration, seed level outputs and generated tables are retained by the simulation code so that the aggregate claims can be traced back to individual streams.

5.2. Metrics and Statistical Reporting

Performance is assessed through event-level detection, delay, task preservation, overhead and reliability metrics. A true positive is recorded when a committed repair occurs at or after the effective drift onset. A false positive is recorded when a repair is committed on a matched null stream. Detection delay is defined by Equation (19).

$$D = \tau_{\text{alarm}} - \tau_s. \quad (19)$$

Three outcome metrics are additionally defined that cannot be supplied by a transmitted class decision, so that the contribution of mismatch detection and codebook repair can be separated from the contribution of the task belief. The residual functional semantic outage rate is defined in Equation (20).

$$\Omega = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} \mathbb{I}[g_\psi(c_{k_t}^{\text{rx}}) \neq g_\psi(c_{k_t}^{\text{tx}})], \quad (20)$$

where $\mathcal{A}$ is the set of accepted post-onset slots and the receiver codebook is restored after an acknowledged repair. The residual outage duration is $\tau_{\text{ack}} - \tau_s$. The semantic reconstruction fidelity is defined in Equation (21).

$$\text{PSNR} = 10 \log_{10} \frac{x_{\text{max}}^2}{\frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} \|x_t - \hat{x}_t^{\text{rx}}\|_2^2 / d_x}, \quad (21)$$

where $\hat{x}_t^{\text{rx}}$ is obtained by inverting the learned projection on the received semantic representation. Post-repair codebook accuracy is measured on the tail after τ_{ack} with the task-belief path disabled. Task accuracy is also reported over accepted packets only, because the four-bit belief is protected by fivefold repetition and a rescue exchange, while the payload uses threefold repetition, so the two paths do not experience the same residual packet error.

The detection probability is treated as event recall. Matched null specificity is $1 - \text{FAR}$. Balanced event accuracy is the mean of recall and specificity. Equal prior event precision is defined as $\text{Recall} / (\text{Recall} + \text{FAR})$ and the F1 score is the harmonic mean of precision and recall. The Matthews correlation coefficient is computed from the event-level true positive, true negative, false positive and false negative counts. Post-onset task accuracy measures

the fraction of correct downstream classifications after the effective drift onset. Charged overhead contains probe messages, resynchronisation messages, the compact task belief header and protected rescue tokens. Control-only overhead is also retained in the ablation so that the task preservation cost can be isolated.

A common network-accounting model is applied to every method in the controlled comparison. Each source sample represents an 8×8 greyscale image and is assigned $b_s = 512$ source-equivalent bits. One sample is offered in each $\Delta = 1$ ms slot. The radio bandwidth is $B = 1$ MHz, the transmit power is $P_{tx} = 0.01$ W (10 mW or 10 dBm) and the repeated-sequence index with cyclic redundancy checking occupies $b_{ch} = 45$ channel uses per payload. The coded task belief is denoted by S_m and is treated as useful semantic side information because it directly supplies the preserved task decision. Protected probes, repair requests, codebook transfer and acknowledgements are denoted by C_m and are treated as control traffic. The task-weighted data rate is defined in Equation (22).

$$R_{w,m} = \frac{b_s}{T\Delta} \sum_{t=1}^T \mathbb{I}[\hat{y}_{m,t} = y_t]. \quad (22)$$

This metric measures the source-equivalent rate for which the downstream task result is correct. The bandwidth-normalised semantic spectral efficiency is defined in Equation (23).

$$\eta_{s,m} = \frac{R_{w,m}}{B}. \quad (23)$$

Because the same 1 MHz bandwidth is used by every method, the ordering of $\eta_{s,m}$ is identical to the ordering of the task-weighted data rate. The control-adjusted semantic resource efficiency is defined separately in Equation (24).

$$\zeta_{s,m} = \frac{b_s \sum_{t=1}^T \mathbb{I}[\hat{y}_{m,t} = y_t]}{Tb_{ch} + S_m + C_m}. \quad (24)$$

This second quantity retains the cost of payload coding, semantic side information and control traffic without mislabelling it as spectral efficiency. The control traffic share is defined in Equation (25).

$$\rho_{c,m} = \frac{C_m}{Tb_{ch} + S_m + C_m}. \quad (25)$$

The radio transmission energy is defined in Equation (26).

$$E_{tx,m} = \frac{P_{tx}}{B} (Tb_{ch} + S_m + C_m). \quad (26)$$

The power in Equation (26) is expressed in watts and bandwidth is expressed in hertz, so the computed energy is in joules and is converted to millijoules for reporting. Only the explicitly modelled radio energy is included. Training, processor and memory energy are not inferred from software execution time. This boundary is stated so that the values are reproducible and are not confused with the joint communication and computation energy used in energy-focused semantic communication studies [33,35].

Uncertainty is reported through confidence intervals selected according to the type of metric. Mean delay, task accuracy and overhead are accompanied by two-sided 95 percent Student intervals. Detection and false alarm proportions are accompanied by exact Clopper-Pearson intervals. A threshold sweep is used to display delay against matched null false alarm rate. Best values in the controlled aggregate comparison are bolded and ties are bolded jointly.

The paired stream design is important for interpretation. Each method is evaluated on the same harmful and null realisations, which reduces variation caused by channel sampling and makes method differences attributable to their response to the same evidence. Conditional delay is reported together with detection coverage because a detector that misses difficult streams can appear artificially fast when delay is averaged only over successful alarms. Zero observed false alarms are accompanied by a nonzero upper confidence bound and are not described as proof of zero population risk.

Network quantities are derived from the same slot-level task outputs and message logs rather than from an independent analytical scenario. Task-weighted data rate and semantic spectral efficiency increase only when more source samples receive a correct downstream decision. Control-adjusted resource efficiency decreases when additional useful side information or control messages consume channel uses. Control share is interpreted jointly with absolute control traffic so that a larger denominator cannot be mistaken for a reduction in transmitted control bits. These definitions preserve the distinction between task benefit, radio resource use and control burden.

5.3. Detector Traces and Null Reliability

Detector operation over time is illustrated through representative harmful and matched-null traces. A representative gradual divergence trace is shown in Figure 2. Global evidence, semantic class local evidence, fused evidence, the learned mismatch score, the evidence and belief gates, the evaluation onset, the protected probe events and the committed repair are displayed with enlarged labels. A matched null trace is shown in Figure 3 under the same fading and interference process without semantic divergence, where the passive evidence and learned mismatch belief remain below their repair conditions.

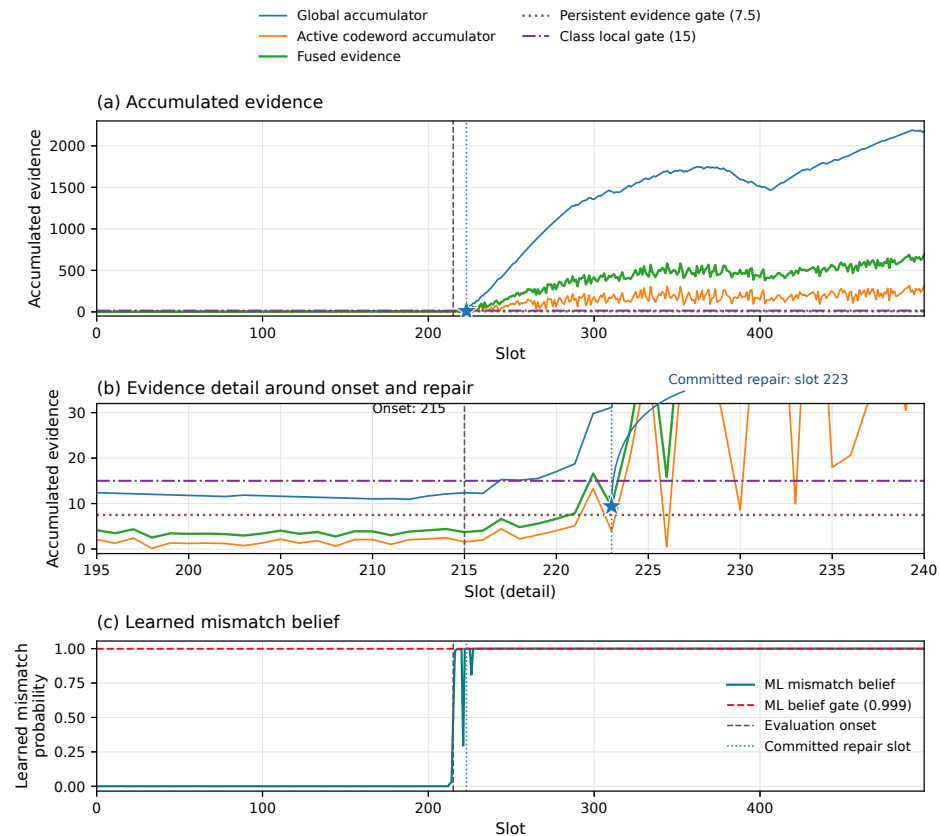


Figure 2. Representative gradual codebook divergence, showing accumulated evidence, an evidence detail around onset with the blue star marking committed repair at slot 223 and the learned mismatch belief.

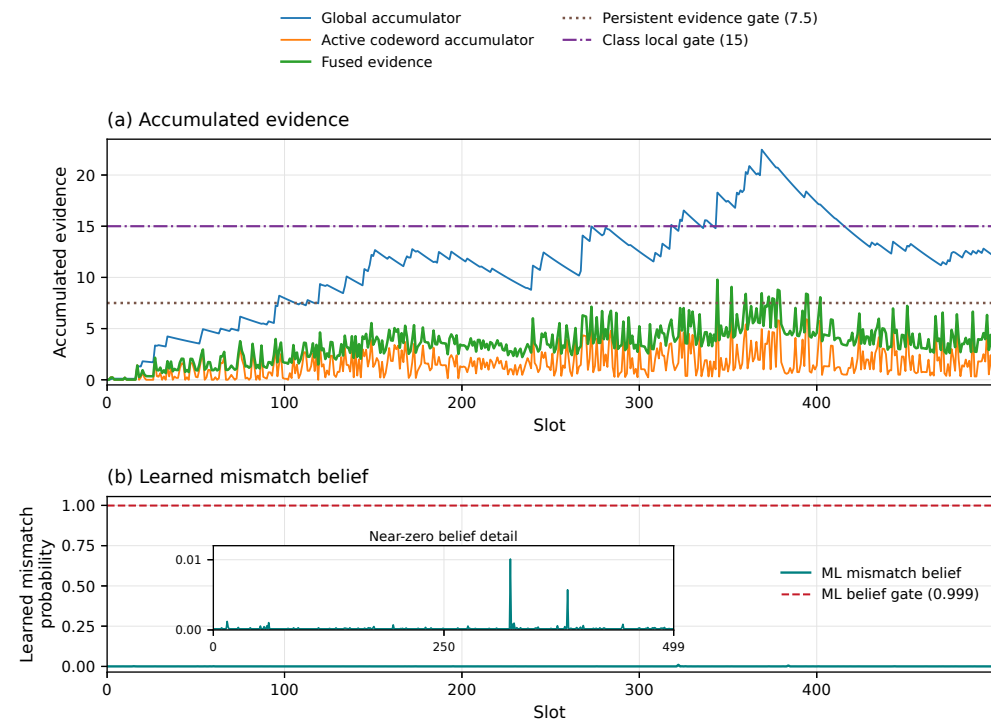


Figure 3. Representative matched null stream under the Rician and interference process used by the harmful streams, with accumulated evidence above and learned mismatch belief below.

Null reliability is reported in Table 4 and Figure 4. CoSMA DAI produces an observed matched null false alarm rate of 0.00 percent with an exact upper 95 percent confidence limit of 1.22 percent. Reference CTM obtains the same observed rate. The result is therefore described as zero observed false alarms rather than perfect reliability. Monitors with shorter aggregate delay accept a nonzero matched null false alarm rate, while the learned CoSMA DAI belief retains the zero observed false alarm operating point.

Table 4. False alarm reliability with exact 95 percent confidence intervals.

Method	Null Channel	Runs	False Alarms	Rate	95% Interval
CoSMA DAI	Matched autoregressive	300	0	0.00%	[0.00%, 1.22%]
	Degrading ramp stress	300	0	0.00%	[0.00%, 1.22%]
CUSUM [11]	Matched autoregressive	300	14	4.67%	[2.57%, 7.71%]
	Degrading ramp stress	300	64	21.33%	[16.83%, 26.41%]
Page Hinkley [12]	Matched autoregressive	300	2	0.67%	[0.08%, 2.39%]
	Degrading ramp stress	300	3	1.00%	[0.21%, 2.89%]
WATCH [16]	Matched autoregressive	300	1	0.33%	[0.01%, 1.84%]
	Degrading ramp stress	300	1	0.33%	[0.01%, 1.84%]
Diagnostic E Process [15]	Matched autoregressive	300	6	2.00%	[0.74%, 4.30%]
	Degrading ramp stress	300	3	1.00%	[0.21%, 2.89%]
Conditional CTM [17]	Matched autoregressive	300	1	0.33%	[0.01%, 1.84%]
	Degrading ramp stress	300	1	0.33%	[0.01%, 1.84%]
Reference CTM [14]	Matched autoregressive	300	0	0.00%	[0.00%, 1.22%]
	Degrading ramp stress	300	0	0.00%	[0.00%, 1.22%]

The degrading ramp control exposes the value of channel conditioning and protected confirmation more clearly than the matched null alone. CUSUM increases from 4.67 percent false alarms on the matched null to 21.33 percent under sustained channel degradation. Page Hinkley reaches 1.00 percent on the ramp and the Diagnostic E Process reaches 1.00 percent. WATCH and Conditional CTM each produce one false alarm in both null

campaigns. CoSMA DAI and Reference CTM produce no observed false alarms in either campaign. The proposed result is therefore supported by two distinct physical controls, although the exact confidence interval still limits the strength of any population claim.

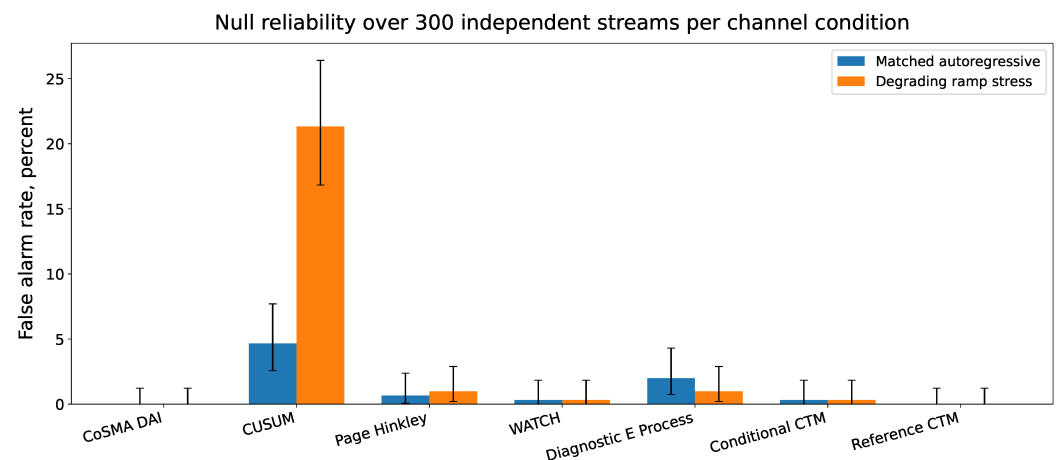


Figure 4. False alarm rates over the matched null and degrading ramp control streams.

The representative traces explain how the zero observed operating point is achieved. Under gradual mismatch, global and local evidence rise after the evaluation onset, a candidate is maintained long enough to request protected confirmation and the mismatch score crosses its selected gate before repair. Under the matched null, score excursions occur because the radio process remains stochastic, but they do not form the combined persistence, class local and probe pattern required by the guarded repair path. Reliability is therefore produced by the sequence of conditioning, memory, active evidence and belief gating rather than by one isolated high threshold.

5.4. Controlled Comparison with State of the Art Monitors

Table 5 provides the controlled quantitative comparison with the referenced monitors. The proposed protocol is compared with CUSUM [11], Page Hinkley [12], WATCH [16], the Diagnostic E Process [15], Conditional CTM [17] and Reference CTM [14]. Every method receives identical channel-conditioned evidence and is evaluated on identical harmful and matched null streams.

The aggregate results are interpreted using delay, event-level reliability, task accuracy and charged overhead. Delay is averaged across valid detections over abrupt, gradual and sparse divergence. False alarm rate is measured over 300 matched null streams. Balanced accuracy, precision, recall, F1 score and Matthews correlation coefficient describe event-level detection under an equal harmful and null prior. Task accuracy measures downstream semantic preservation and overhead includes every charged control and task-belief bit. In Table 5, delay is reported in slots, false alarm rate and classification metrics are reported in percent and overhead is reported in kbit per harmful stream. Best values are bolded and ties are bolded jointly. Table 5 shows that CoSMA DAI obtains 100.00 percent balanced accuracy, precision, recall, F1 score and Matthews correlation coefficient with zero observed matched null false alarms. Reference CTM ties these reliability and classification values. CoSMA DAI reduces aggregate delay from 11.58 slots to 5.62 slots relative to Reference CTM, which is a 51.44 percent reduction and makes the proposed protocol the fastest evaluated method at the zero observed false alarm operating point. The task accuracy of 94.73 percent is maintained in every scenario by the task-preservation belief and is a property of that mechanism; it is not used to separate the detectors, because a transmitted class decision reproduces the same value with no detection and no repair, as shown in

Section 5.5. The reported cost is not free. Total charged overhead rises to 16.81 kbit, of which 9.29 kbit is associated with probing and resynchronisation and 7.52 kbit is associated with task belief preservation. The proposed method therefore provides the strongest reliability constrained detection result, while lower communication cost remains available from the baseline monitors.

The aggregate table also reveals why raw delay alone is insufficient. CUSUM is the fastest monitor at 2.04 slots, although its recall falls to 90.00 percent and its matched null false alarm rate rises to 4.67 percent. Page Hinkley, WATCH and Conditional CTM are faster than CoSMA DAI but accept nonzero false alarms. Reference CTM reaches the same event classification values and zero observed false alarm rate as CoSMA DAI, although its 11.58 slot delay leaves the receiver exposed to the divergent semantic mapping for approximately twice as long. The proposed advantage is therefore located on the reliability constrained frontier rather than at the unconstrained minimum delay point.

The task accuracy column of Table 5 shows what task preservation achieves. The monitor baselines depend on the receiver representation until their alarm causes repair, so their task accuracy remains between 78.05 percent and 84.19 percent, whereas the authenticated transmitter decision keeps the application at 94.73 percent throughout the confirmation and repair interval, above the 0.919 ceiling that a perfectly aligned semantic path can reach, because the belief is formed from the unquantised latent. Section 5.5 shows that the same value is obtained by transmitting the class decision alone and that every monitor reaches it once the same header is supplied, so the column measures the preservation mechanism rather than the detector and the detectors are separated on residual functional semantic outage, outage duration and reconstruction fidelity.

Table 5. Controlled aggregate comparison with referenced state-of-the-art sequential monitors. The six monitors are the compact implementations described in Section 5.1; the full paper-defined algorithms are reported separately in this subsection.

Method	Delay ↓	FAR ↓	Bal. Acc. ↑	Precision ↑	Recall ↑	F1 ↑	MCC ↑	Task Acc. ↑	Overhead ↓
CoSMA DAI (proposed)	5.62	0.00	100.00	100.00	100.00	100.00	100.00	94.73	16.81
CUSUM [11]	2.04	4.67	92.67	95.07	90.00	92.47	85.45	78.05	8.32
Page Hinkley [12]	3.39	0.67	99.67	99.34	100.00	99.67	99.34	84.19	8.32
WATCH [16]	3.84	0.33	99.83	99.67	100.00	99.83	99.67	84.17	8.32
Diagnostic E Process [15]	7.87	2.00	99.00	98.04	100.00	99.01	98.02	83.72	8.32
Conditional CTM [17]	3.60	0.33	99.83	99.67	100.00	99.83	99.67	84.19	8.32
Reference CTM [14]	11.58	0.00	100.00	100.00	100.00	100.00	100.00	83.55	8.32

Notes: ↑ indicates that a larger value is preferred; ↓ indicates that a smaller value is preferred. Bold numeric values identify the best result in each column, including ties.

Scenario-specific results are reported in Table 6. CoSMA DAI obtains delays of 2.33, 4.97 and 9.57 slots under abrupt, gradual and sparse divergence. The corresponding retained task accuracies are 0.947, 0.947 and 0.947 in the three scenarios, which shows that the preservation belief is insensitive to the divergence geometry; as Section 5.5 establishes, these values measure the preservation mechanism rather than the detector. Detection coverage is shown beside conditional delay so that a low delay cannot conceal missed streams.

Abrupt divergence produces the shortest CoSMA DAI delay because broad evidence rises immediately, while the protected probe still prevents the near-zero-delay response observed for CUSUM from being interpreted as confirmed repair. Gradual divergence increases the proposed delay to 4.97 slots because the evidence must accumulate after the interpolation reaches the declared onset. Sparse divergence is the most difficult case and increases delay to 9.57 slots because affected classes appear intermittently. The class local route nevertheless retains 100 percent detection and 0.947 task accuracy. Reference CTM requires 24.07 slots in the sparse case and the Diagnostic E Process requires 10.33 slots,

while the faster Page Hinkley, WATCH and Conditional CTM solutions continue to provide approximately 0.841 task accuracy because no independent task preservation path is available.

Table 6. Scenario specific held out results over 30 seeds.

Method	Scenario	Detection Rate	Delay, Slots	Task Accuracy	Overhead, kbit
CoSMA DAI (proposed)	Abrupt	100.0%	2.33 [1.43, 3.24]	0.947 [0.943, 0.952]	16.73
	Gradual	100.0%	4.97 [3.85, 6.09]	0.947 [0.943, 0.952]	16.63
	Sparse	100.0%	9.57 [5.07, 14.06]	0.947 [0.943, 0.952]	17.07
CUSUM [11]	Abrupt	90.0%	0.04 [0.00, 0.11]	0.758 [0.662, 0.854]	8.32
	Gradual	90.0%	2.85 [2.00, 3.71]	0.767 [0.678, 0.856]	8.32
	Sparse	90.0%	3.22 [1.86, 4.59]	0.817 [0.786, 0.847]	8.32
Page Hinkley [12]	Abrupt	100.0%	0.30 [0.06, 0.54]	0.841 [0.831, 0.851]	8.32
	Gradual	100.0%	5.40 [4.03, 6.77]	0.844 [0.834, 0.854]	8.32
	Sparse	100.0%	4.47 [2.93, 6.00]	0.841 [0.831, 0.851]	8.32
WATCH [16]	Abrupt	100.0%	0.37 [0.12, 0.62]	0.841 [0.831, 0.850]	8.32
	Gradual	100.0%	6.57 [4.83, 8.31]	0.844 [0.834, 0.854]	8.32
	Sparse	100.0%	4.60 [3.02, 6.18]	0.841 [0.831, 0.851]	8.32
Diagnostic E Process [15]	Abrupt	100.0%	3.10 [2.34, 3.86]	0.832 [0.822, 0.842]	8.32
	Gradual	100.0%	10.17 [7.99, 12.35]	0.842 [0.832, 0.852]	8.32
	Sparse	100.0%	10.33 [7.73, 12.93]	0.837 [0.827, 0.847]	8.32
Conditional CTM [17]	Abrupt	100.0%	0.30 [0.06, 0.54]	0.841 [0.831, 0.851]	8.32
	Gradual	100.0%	6.03 [4.42, 7.64]	0.844 [0.834, 0.854]	8.32
	Sparse	100.0%	4.47 [2.93, 6.00]	0.841 [0.831, 0.851]	8.32
Reference CTM [14]	Abrupt	100.0%	1.53 [1.23, 1.84]	0.837 [0.828, 0.847]	8.32
	Gradual	100.0%	9.13 [7.05, 11.21]	0.843 [0.833, 0.853]	8.32
	Sparse	100.0%	24.07 [15.77, 32.36]	0.827 [0.814, 0.839]	8.32

Notes: Bold numeric values identify the best result within each scenario and metric, including ties. Higher detection rate and task accuracy are preferred; lower delay and overhead are preferred. Brackets give 95% confidence intervals.

Delay intervals overlap for some monitors and scenarios, so universal delay superiority is not claimed. The scenario results support three narrower conclusions: full detection is retained across abrupt, gradual and sparse mismatch, the residual outage interval is bounded in every scenario and the protected confirmation cost becomes most visible when the affected semantic classes are observed infrequently.

Figures 5 and 6 visualise detection delay and task accuracy. The accuracy axis begins at zero so that small differences are not exaggerated. Error bars represent 95 percent confidence intervals. Larger type, reduced label rotation and additional spacing are used so that legends, tick labels and titles do not collide.

Table 7 and Figure 7 provide the controlled network-metric comparison. The comparison uses the same learned semantic source, physical channel, stream duration and held-out seeds for CoSMA DAI and the six referenced monitors. Table 7 reports means with 95 percent confidence intervals over 90 held-out harmful streams. Semantic spectral efficiency is the task-weighted useful rate divided by the common bandwidth. Control-adjusted efficiency includes payload coding, semantic side information and control traffic. Control share includes probes, repair and acknowledgements but excludes the useful task-belief side information. Radio energy includes the explicitly modelled ten dBm transmitter only. Best values are bolded and ties are bolded jointly. The energy-oriented and semantic-rate studies reviewed earlier [10,33–35] motivate the selected metrics, although their published numerical values are not transferred because their tasks and resource models differ.

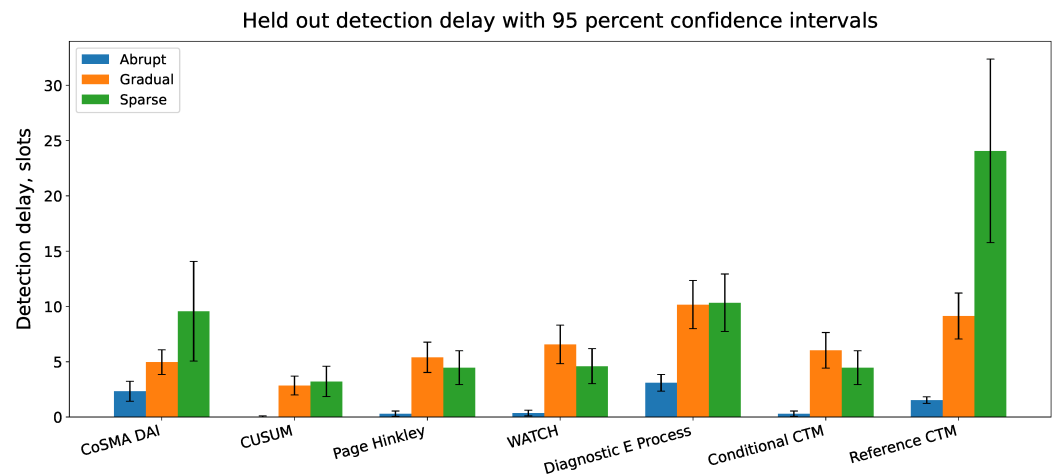


Figure 5. Detection delay with 95 percent confidence intervals for abrupt, gradual and sparse semantic divergence.

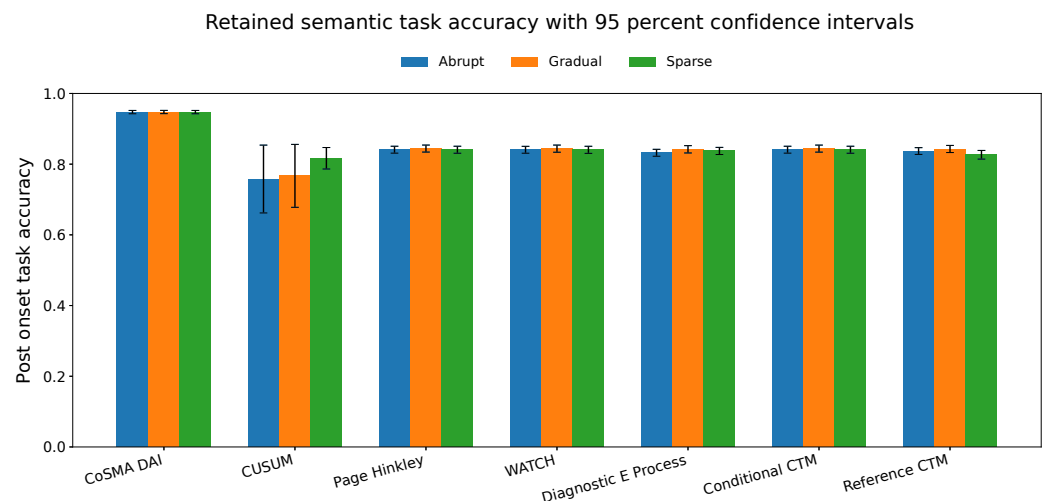


Figure 6. Post-onset task accuracy with 95 percent confidence intervals.

Table 7. Controlled network-metric comparison under the common physical accounting model.

Method	Task-Weighted Rate $\uparrow$ Mbit/s	Semantic SE $\uparrow$ bit/s/Hz	Control-Adjusted Efficiency $\uparrow$ Source bit/Channel Use	Radio Energy $\downarrow$ mJ/Stream	Control Traffic $\downarrow$ kbit/Stream	Control Share $\downarrow$ Percent
CoSMA DAI (proposed)	0.487 [0.486, 0.489]	0.487 [0.486, 0.489]	6.203 [6.168, 6.239]	0.393 [0.391, 0.395]	9.29 [9.10, 9.47]	23.59 [23.24, 23.93]
CUSUM [11]	0.414 [0.400, 0.427]	0.414 [0.400, 0.427]	6.711 [6.493, 6.930]	0.308 [0.308, 0.308]	8.32 [8.32, 8.32]	27.00 [27.00, 27.00]
Page Hinkley [12]	0.432 [0.430, 0.434]	0.432 [0.430, 0.434]	7.005 [6.969, 7.042]	0.308 [0.308, 0.308]	8.32 [8.32, 8.32]	27.00 [27.00, 27.00]
WATCH [16]	0.432 [0.429, 0.434]	0.432 [0.429, 0.434]	7.004 [6.967, 7.041]	0.308 [0.308, 0.308]	8.32 [8.32, 8.32]	27.00 [27.00, 27.00]
Diagnostic E Process [15]	0.430 [0.428, 0.433]	0.430 [0.428, 0.433]	6.982 [6.945, 7.020]	0.308 [0.308, 0.308]	8.32 [8.32, 8.32]	27.00 [27.00, 27.00]
Conditional CTM [17]	0.432 [0.430, 0.434]	0.432 [0.430, 0.434]	7.005 [6.968, 7.042]	0.308 [0.308, 0.308]	8.32 [8.32, 8.32]	27.00 [27.00, 27.00]
Reference CTM [14]	0.430 [0.428, 0.432]	0.430 [0.428, 0.432]	6.975 [6.935, 7.014]	0.308 [0.308, 0.308]	8.32 [8.32, 8.32]	27.00 [27.00, 27.00]

Notes: $\uparrow$ indicates that a larger value is preferred; $\downarrow$ indicates that a smaller value is preferred. Bold numeric values identify the best displayed mean in each column, including ties. Brackets give 95% confidence intervals.

The network results show how task preservation changes useful rate, spectral efficiency, control share and energy. CoSMA DAI reaches a task-weighted data rate of 0.487 Mbit/s, compared with 0.432 Mbit/s for the strongest comparator. The corresponding gain is 12.88 percent and is produced by the higher number of correctly served task samples. The bandwidth-normalised semantic spectral efficiency is 0.487 bit/s/Hz, compared with 0.432 bit/s/Hz, which gives the same 12.88 percent gain under the common 1 MHz bandwidth. The correctly partitioned control traffic share is 23.59 percent, compared with 27.00 percent for the monitors and is reduced by 12.62 percent. The absolute

control traffic remains 9.29 kbit per stream because protected probes are added to one repair exchange. The lower share is therefore not interpreted as lower absolute control traffic. The control-adjusted semantic resource efficiency is 6.203 source-equivalent bit per channel use and the radio energy is 0.393 mJ per stream. Monitors without task-belief side information retain higher control-adjusted resource efficiency and lower radio energy, so this trade-off remains visible.

Controlled network-metric comparison on identical held-out streams

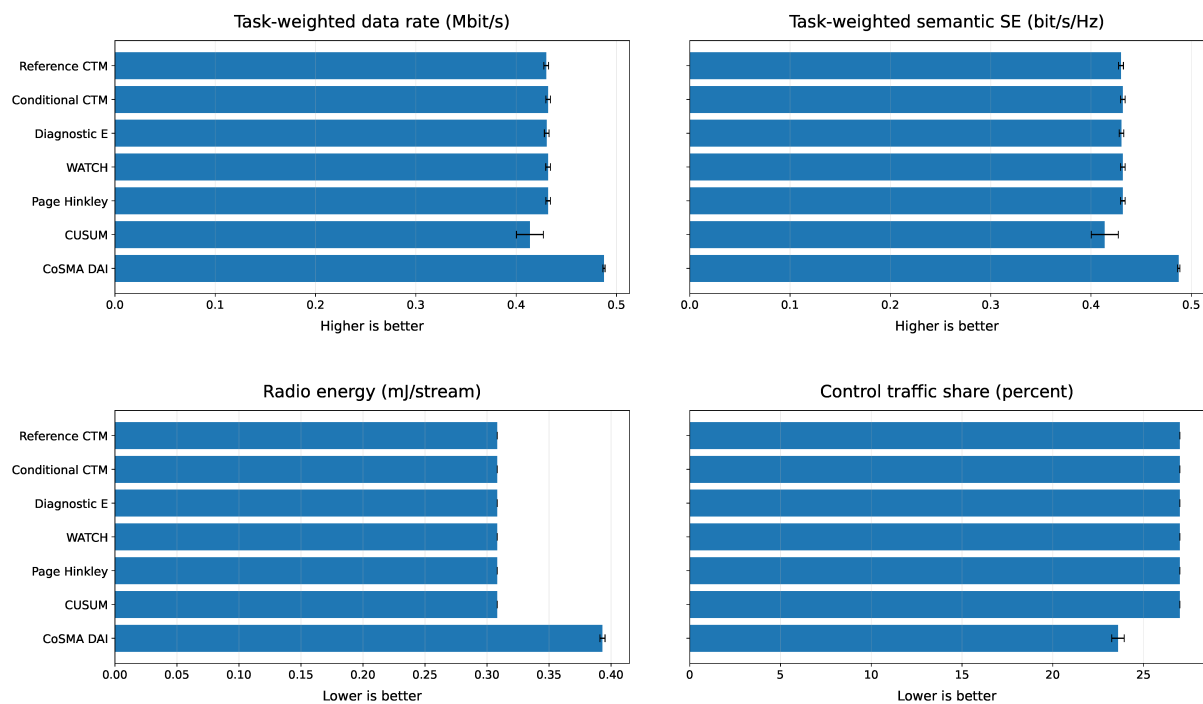


Figure 7. Task-weighted data rate, semantic spectral efficiency, radio energy consumption and control-traffic share under identical held-out streams.

The rate gain and the control share reduction arise from different parts of the accounting. The rate gain is produced by additional correct task decisions after semantic divergence. The control share reduction is produced because useful task belief bits are placed in the semantic side information term rather than in the probe and repair control numerator. Those bits are not removed from the denominator and remain charged in radio energy. Consequently, the proposed control share of 23.59 percent can be lower than the 27.00 percent monitor value even though absolute control traffic rises to 9.29 kbit and total radio energy rises to 0.393 mJ per stream.

The control-adjusted resource efficiency exposes the cost of protecting the task. Page Hinkley and Conditional CTM reach approximately 7.005 source-equivalent bits per channel use, while CoSMA DAI reaches 6.203. The proposed method therefore converts more source samples into correct task outcomes but spends additional channel uses on authenticated task beliefs and protected confirmation. The result should be read as a reliability and task preservation trade-off rather than as universal resource dominance. Applications with inexpensive errors may prefer the lower energy monitor operating points, while applications that require zero observed false repairs and sustained task correctness obtain the stronger result from CoSMA DAI.

Table 8 provides a source-grounded comparison with recent semantic communication studies. The table records the network metric defined by each source and the result claimed

under its own task and resource model. The source-reported outcomes are retained in their original evaluation domains and are not treated as directly interchangeable numerical measurements. Numerical values from different source models are not placed on the same scale because such a transfer would mix incompatible semantic utilities, bandwidth definitions, computation boundaries and channel assumptions. The last row reports the controlled CoSMA DAI measurements, while Table 7 remains the directly comparable numerical experiment.

Table 8. Source-grounded comparison of network efficiency metrics used by recent semantic communication approaches.

Approach	Rate or Spectral Metric	Energy Boundary	Source-Reported Result	Relation to the Controlled CoSMA DAI Comparison
Energy-efficient semantic RSMA [33]	Semantic extraction rate with common and private rate-splitting resources	Joint computation and wireless transmission energy	Joint semantic extraction, computation frequency and radio allocation reduced total energy under latency and power constraints	Motivates joint energy accounting. Its task, RSMA transmission and computation model differ from the present held-out drift streams, so its source values are not copied into Table 7.
On-demand semantic forwarding [34]	Semantic energy efficiency based on recovered semantic utility per transmit energy	Source, relay and destination transmission energy	Almost the same recovery as always-on forwarding was reported with lower energy through demand-triggered relay use	Supports event-triggered control. The relay topology and semantic similarity metric differ from the present direct mobile link.
Aerial-aided green semantic communication [35]	Spectrum-efficiency and service-revenue objectives for aerial edge service	Communication, semantic processing and distributed model-update energy	The proposed incentive and distributed learning mechanisms improved energy efficiency, training accuracy and training energy consumption	Demonstrates that computation and learning energy can dominate in edge networks. The present result reports radio energy separately and states this boundary explicitly.
Knowledge-sharing semantic rate maximisation [10]	Generalised effective semantic transmission rate combining semantic accuracy and knowledge-sharing overhead	Radio and knowledge-sharing resource constraints	Higher effective semantic rate was reported, especially at low signal-to-noise ratio, through joint sharing, extraction and association decisions	Closest conceptual rate comparison. CoSMA DAI uses a task-weighted rate and accounts separately for task-belief side information, probes and repair traffic under one fixed physical model.
MOC-RVQ digital semantic coding [26]	Channel-bandwidth ratio for a multilevel residual codebook	Transmission bandwidth; energy was not the primary reported boundary	Comparable performance to analogue joint source-channel coding was reported with approximately one sixth of the channel-bandwidth ratio	Supports digital codebook compression but does not evaluate online mismatch confirmation or resynchronisation energy.
CoSMA DAI (proposed)	Task-weighted rate: 0.487 Mbit/s; semantic spectral efficiency: 0.487 bit/s/Hz; control-adjusted efficiency: 6.203 source-equivalent bit/channel use	Radio energy: 0.393 mJ/stream at 10 dBm; semantic task-belief traffic and control traffic are accounted for separately	Highest task-weighted rate and semantic spectral efficiency in the controlled comparison, with a 12.88 percent gain; the control share is 23.59 percent and absolute control traffic and radio energy are disclosed	Directly measured against six referenced monitors on identical streams in Table 7; computation energy is excluded and identified as a limitation.

Figure 8 compares the operating characteristics across the evaluated threshold range. Delay is plotted against the matched null false alarm rate across a threshold sweep. The learned belief gate controls the proposed reliability and response time balance. A lower gate produces a delay of 3.64 slots with a 3.3 percent false alarm rate, while the selected gate produces a delay of 5.62 slots with zero observed false alarms. The ordering changes with the reliability requirement, although CoSMA DAI provides the lowest delay among the evaluated zero-observed false-alarm operating points.

The sweep also shows that the selected operating point is a design choice rather than an unavoidable property of the estimator. Reducing the belief gate to 0.995 moves the method toward the faster monitor region but introduces a 3.3 percent false alarm rate. Increasing the gate further preserves the zero observed false alarm result but adds delay. The selected value of 0.999 was retained because the study prioritises false repair avoidance before minimum response time. A different application can choose another point from the same learned belief output without changing the evidence construction or the BDIX plan library.

The compact monitor implementations of the principal comparison are complemented in Table 9 by the full algorithms as published, obtained from an independent implementation audit in which each recursion was written from the source paper and checked against the authors' public code where it exists. The full WATCH, Conditional CTM and Reference CTM constructions detect every event but with delays of 42.82 to 50.66 slots on the principal streams and the full diagnostic martingales detect 68.9 percent of events, because their exchangeability and reference assumptions are not met by a correlated fading link and a score with a point mass at zero. The compact implementations are therefore stronger comparators on this benchmark than the published algorithms they follow and the principal comparison is conservative in that sense. The two tables are reported side by side rather than merged, because the full algorithms are calibrated to an empirical one percent false-alarm target on separate null streams, whereas the compact implementations use their development-selected gates, so the two operating points are not interchangeable.

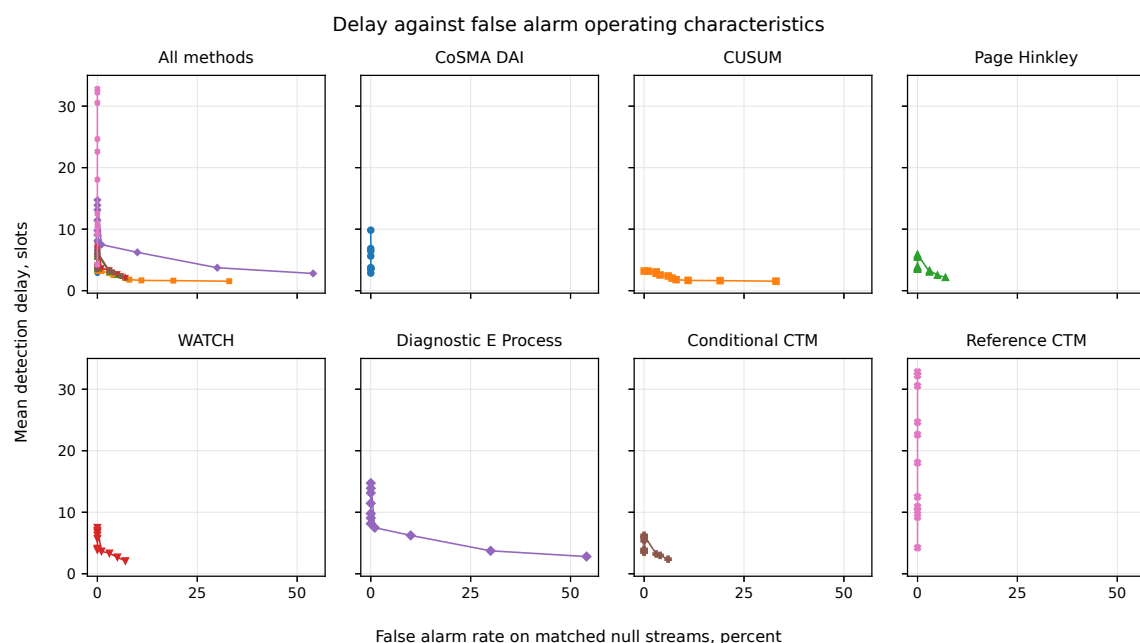


Figure 8. Detection delay against false alarm rate across a threshold sweep, with an overview and separate method panels on identical scales to expose coincident operating points.

Table 9. Full paper-defined implementations of the six monitors on the principal streams, from an independent implementation audit. Each monitor is calibrated on 300 separate null streams to target an empirical one percent probability of any alarm within a stream. The false-alarm column reports the observed rate on held-out null streams; these rates can differ from the calibration target and between monitors. Restricted delay assigns the remaining horizon to a missed event. The principal comparison of Table 5 uses the compact implementations at their development-selected gates and is not superseded by this table; the two are reported together so that the reader may judge the effect of implementation fidelity.

Monitor	Recall %	Delay	Restricted Delay	FAP %	95% Interval
CUSUM (standard recursion)	100.0	3.20	3.20	1.00	[0.21, 2.89]
Page Hinkley (standard recursion)	100.0	3.51	3.51	0.67	[0.08, 2.39]
WATCH, full (covariate CTM, density-ratio weights, composite jumper)	100.0	49.72	49.72	1.67	[0.54, 3.85]
Diagnostic martingales, full (recency predictor, Bernoulli wealth)	68.9	131.44	180.77	0.67	[0.08, 2.39]
Conditional CTM, full (fixed CDF, DKW correction, ONS betting)	98.9	50.66	53.32	2.00	[0.74, 4.30]
Reference CTM, full (randomised conformal ranks, Simple Jumper)	100.0	42.82	42.82	2.67	[1.16, 5.19]

5.5. Task Label Substitution Audit and Equal Resource Baselines

A closed-set task metric can be satisfied by transmitting the task decision itself, so a task-label-only control is evaluated in which the four-bit transmitter decision and its rescue token are transmitted with the identical coding, while no evidence is accumulated, no probe is issued and no codebook repair is performed. A semantic-payload-without-repair control and an onset oracle with the label channel disabled bound the comparison and every sequential monitor is additionally evaluated with the identical task-belief header and fusion rule so that no method is advantaged by the side information. Table 10 and Figure 9 report the outcome.

Table 10. Task-label substitution audit and equal-resource baselines on identical held-out streams. Task accuracy depends on the label channel, whereas residual functional semantic outage, post-repair codebook accuracy and reconstruction fidelity depend only on semantic repair. The quantised codebook ceiling is 0.919.

Method	Det. %	FAR %	Delay Slots	Task Acc.	Outage Rate	Outage Slots	Recon. dB
Task label only	0.0	0.00	N/A	0.947	0.7369	288.3	15.97
Semantic payload, no repair	0.0	0.00	N/A	0.231	0.7369	288.3	15.97
Oracle repair, no label	100.0	0.00	0.00	0.843	0.0016	1.0	21.83
Page Hinkley	100.0	0.67	3.39	0.842	0.0034	4.4	21.80
Reference CTM	100.0	0.00	11.58	0.836	0.0120	12.6	21.69
Page Hinkley + task belief	100.0	0.67	3.39	0.947	0.0034	4.4	21.80
Reference CTM + task belief	100.0	0.00	11.58	0.947	0.0120	12.6	21.69
CoSMA DAI (always-on label)	100.0	0.00	5.62	0.947	0.0073	6.6	21.74
CoSMA DAI (windowed label)	100.0	0.00	5.62	0.846	0.0073	6.6	21.74
CoSMA DAI (no label)	100.0	0.00	5.62	0.839	0.0073	6.6	21.74

N/A indicates that detection delay is not applicable to the task-label-only and no-repair controls, which perform no mismatch detection or repair.

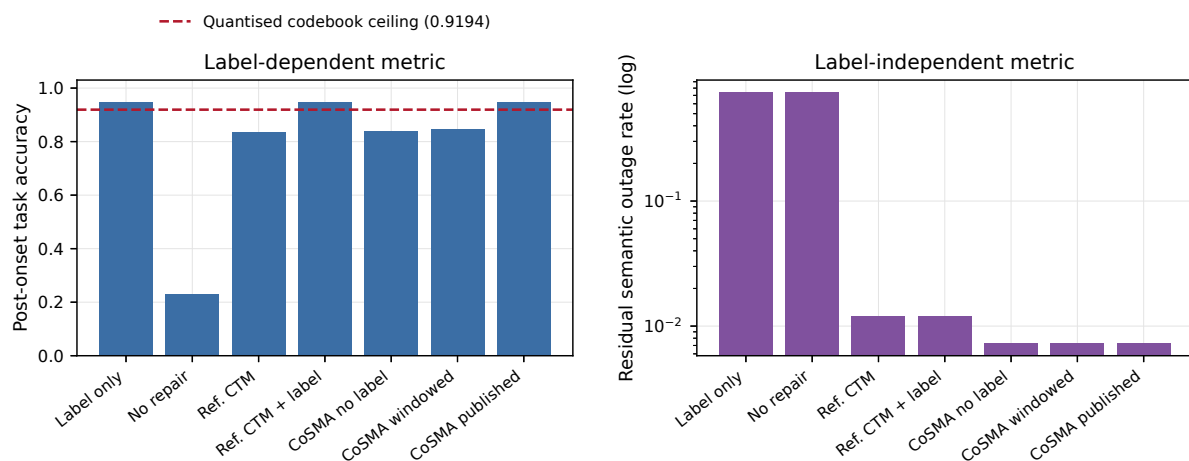


Figure 9. Label-dependent and label-independent outcomes on identical held-out streams, where the red dashed line marks the quantised codebook task-accuracy ceiling of 0.9194 and residual semantic outage distinguishes methods that repair the representation.

The control reaches 94.73 percent post-onset task accuracy, which equals the value obtained by the complete protocol and exceeds the quantised codebook ceiling of 0.919. A value above that ceiling cannot be produced by the semantic path, because the transmitter posterior is formed from the unquantised latent representation rather than from the received codeword. Once the same header is supplied to the comparators, no method retains a task-accuracy advantage. The same control leaves 73.69 percent residual functional semantic outage for the remainder of the stream and 15.97 dB reconstruction fidelity, against 0.73 percent, 6.62 slots and 21.74 dB for the proposed protocol.

The audit therefore separates two distinct functions that a task-accuracy metric alone combines. The task belief preserves the current application decision at 94.73 percent through every scenario, which is its purpose and is achieved in full; the same preservation is available to any detector that carries the header, so it is a property of the mechanism rather than a point of comparison between detectors. Mismatch detection and codebook repair restore the semantic representation itself, which no class decision can do and only this second function is measurable on the label-independent metrics. A variant in which the belief is confined to the interval between the candidate slot and the acknowledged repair is included in the table for information, reaching 84.58 percent task accuracy at 9.44 kbit against 83.88 percent without any belief. For completeness, the table also reports the onset oracle, which bounds the benefit available from detection timing alone.

5.6. Comparison Parity Tiers

The principal comparison supplies one conditioned scalar to the sequential monitors, while the proposed method uses a multivariate causal feature vector, supervised training and active probe evidence. Five parity tiers are therefore declared and reported in Table 11 and Figure 10. Tier A retains the scalar monitors of Section 5.4. Tier B adds Hotelling T^2 and a multivariate exponentially weighted moving average on the identical twenty-dimensional feature vector. Tier C adds logistic regression, a random forest and a multilayer perceptron trained on the identical 190,000 development rows and the identical labels. Tier D supplies the identical protected probe and channel guard to the tier C detectors and to every monitor. Tier E is the proposed method. Every tier receives 144 development candidates and is frozen before any held-out stream is processed.

Table 11. Comparison-parity tiers. Tier A uses one conditioned scalar, tier B uses the identical twenty-dimensional feature vector without supervision, tier C adds the identical supervised training seeds and labels, tier D adds the identical protected probe and channel guard and tier E is the proposed method. Every tier receives 144 development candidates and is frozen before testing.

Tier	Method	Det. %	FAR %	Delay	Outage Rate	Outage Slots
A	CUSUM	90.0	4.67	2.04	0.0774	31.6
A	Page Hinkley	100.0	0.67	3.39	0.0034	4.4
A	Conditional CTM	100.0	0.33	3.60	0.0034	4.6
A	WATCH	100.0	0.33	3.84	0.0038	4.8
A	Diagnostic E Process	100.0	2.00	7.87	0.0091	8.9
A	Reference CTM	100.0	0.00	11.58	0.0120	12.6
B	MEWMA	100.0	0.33	5.73	0.0071	6.7
B	Hotelling T2	96.7	0.67	6.20	0.0308	16.6
C	Multilayer perceptron	98.9	0.33	1.94	0.0132	6.1
C	Random forest	100.0	0.00	8.03	0.0091	9.0
C	Logistic regression	100.0	0.00	21.53	0.0246	22.5
D	Multilayer perceptron + probe	98.9	0.00	2.04	0.0134	6.2
D	CUSUM + probe	96.7	1.33	2.59	0.0283	13.1
D	Logistic regression + probe	100.0	1.00	2.62	0.0041	3.6
D	Page Hinkley + probe	100.0	0.33	3.40	0.0034	4.4
D	Conditional CTM + probe	100.0	0.33	3.60	0.0034	4.6
D	WATCH + probe	100.0	0.33	3.84	0.0038	4.8
D	Random forest + probe	100.0	0.00	4.14	0.0044	5.1
D	Diagnostic E Process + probe	100.0	0.00	7.88	0.0091	8.9
D	Reference CTM + probe	100.0	0.00	11.98	0.0125	13.0
E	CoSMA DAI	100.0	0.00	5.62	0.0073	6.6

Two findings follow from these tiers. First, the predictive advantage does not belong to the selected estimator. Under full parity, Random forest + probe reaches 4.14 slots with zero observed false repairs and 0.44 percent residual outage, against 5.62 slots and 0.73 percent for the proposed histogram gradient boosting belief. No estimator-specific

detection advantage is therefore attributed to the learned belief. Second, the protected confirmation stage improves every detector to which it is attached. The matched null false repair rate of CUSUM falls from 4.67 to 1.33 percent, that of the Diagnostic E Process falls from 2.00 to 0.00 percent, that of the multilayer perceptron falls from 0.33 to 0.00 percent and the delay of the logistic-regression detector falls from 21.53 to 2.62 slots.

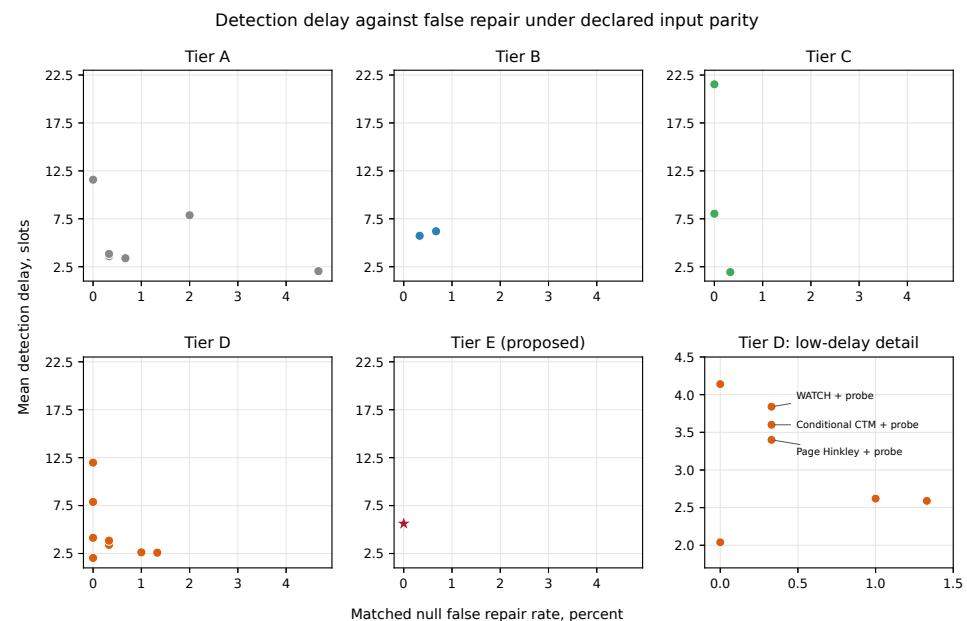


Figure 10. Detection delay against matched null false repair rate under declared input, supervision and probe parity, with separate tier panels on identical scales and a low-delay detail for Tier D.

The contribution of this work is therefore located in the staged confirmation and guard structure rather than in a superior change detector. This interpretation is consistent with the independently written cascade of Section 5.7, which reproduces the controller exactly and shows that the deliberation layer supplies auditability rather than additional discrimination. It is also consistent with the ablation result that removing the protected probe produces the largest false repair rate of any evaluated variant.

5.7. Ablation, Sensitivity and Guarded Plan Results

Table 12 and Figure 11 present the component ablation results. Replacement of the learned mismatch belief updater by one fixed probe threshold reduces mean delay to 2.95 slots, although detection falls to 96.7 percent and false repairs increase to 6.7 percent. Removal of the protected probe reduces delay to 2.49 slots, although detection falls to 90.0 percent and false repairs increase to 30.0 percent. Removal of the task preservation belief leaves detection unchanged, reduces charged overhead from 16.81 kbit to 9.29 kbit and reduces aggregate task accuracy from 0.947 to 0.839. This 10.8 percentage point difference is the size of the label channel itself and is not evidence of a detection benefit, as established in Section 5.5; the residual functional semantic outage is unchanged by this ablation. Global evidence alone retains zero observed false alarms, although delay increases to 8.61 slots. Removal of channel conditioning increases delay to 25.93 slots. These results show that the mismatch belief, task belief, protected confirmation, semantic class local evidence and channel conditioning provide complementary benefits.

The ablations distinguish predictive discrimination from protocol safeguards. Replacing the learned updater with a fixed probe threshold makes the method faster, but lowers detection and introduces false repairs, which shows that heterogeneous passive and active evidence cannot be reduced safely to one probe value. Removing the protected probe

produces the fastest proposed variant but also the largest false alarm rate, confirming that active evidence is the principal repair safeguard. Removing the task belief leaves every detector metric and the residual semantic outage unchanged while reducing task accuracy sharply, which shows that the two functions are independent and that task accuracy alone cannot evaluate semantic repair. Removing local evidence or channel conditioning affects delay rather than only task output, confirming that evidence representation and task preservation act at different stages of the protocol.

The fixed guarded plan library is also compared with an equivalent threshold cascade after the learned belief has been supplied. The cascade row of Table 12 must be read with a caveat: in the simulation code it invokes the same routine as the full protocol with the same arguments, so that row is an alias rather than an ablation and carries no evidence on its own. The equivalence is instead established by a cascade written independently from the decision rules of Section 4.5, without calling the protocol routine, which reproduces the alarm slot and the complete probe schedule of the original controller on 540 of 540 held-out, matched-null and unseen-family streams. This supports the reading that the BDIx layer contributes interpretability, decision structure and the guard behaviour quantified in Section 5.10 rather than additional discriminative power.

Table 12. Component ablation on identical held-out streams.

Variant	Detection Rate	Delay	Accuracy	False Alarm Rate	Overhead, kbit
Full ML belief protocol	100.0%	5.62	0.947	0.0%	16.81
Fixed probe threshold, no ML belief	96.7%	2.95	0.947	6.7%	16.50
No protected probe	90.0%	2.49	0.947	30.0%	16.36
No task preservation belief	100.0%	5.62	0.839	0.0%	9.29
Plain threshold cascade	100.0%	5.62	0.947	0.0%	16.81
Global evidence only	100.0%	8.61	0.947	0.0%	30.52
Unconditioned score	100.0%	25.93	0.947	0.0%	17.10

Notes: Bold numeric values identify the best result in each column, including ties. Higher detection rate and accuracy are preferred; lower delay, false alarm rate and overhead are preferred.

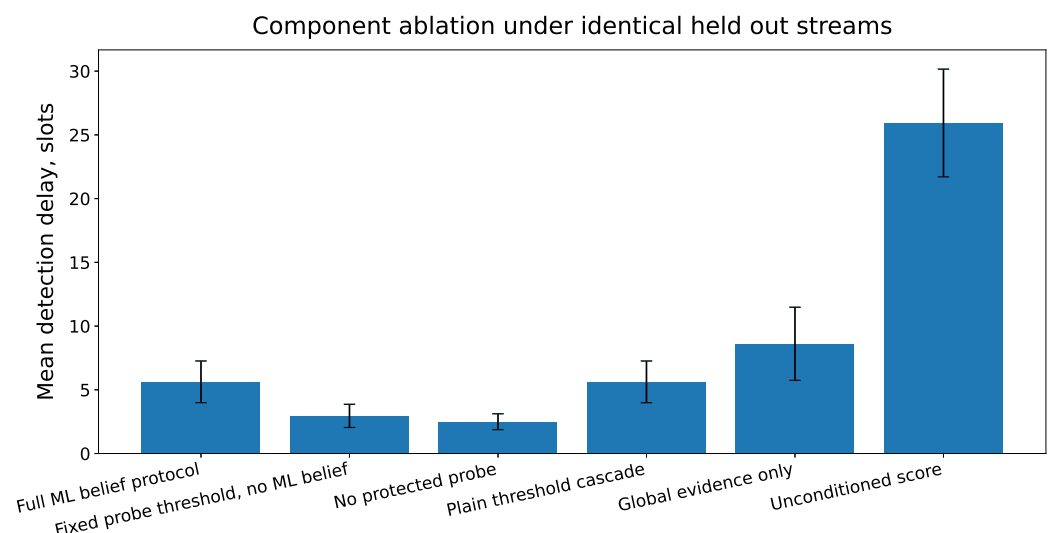


Figure 11. Mean delay and 95 percent intervals for the principal CoSMA DAI ablations.

Sensitivity to the persistent evidence gate, semantic-class local gate, learned-belief gate and evidence persistence is reported in Table 13. The selected belief gate of 0.999 preserves zero observed false alarms with a mean delay of 5.62 slots. Reducing the gate to 0.995 shortens the delay to 3.64 slots, while task accuracy remains protected by the independent task belief and the false alarm rate rises to 3.3 percent. Increasing the gate to 0.9995 preserves

zero observed false alarms and increases delay to 6.86 slots. The evidence gate and local gate alternatives remain within a narrow zero false alarm delay range, which shows that the principal reliability result is not produced by one isolated passive threshold.

Sensitivity is concentrated most strongly in the learned belief gate. Changing the persistent evidence gate from 7.5 to 13.0 and the local gate from 12.0 to 18.0 retains 100 percent detection, zero observed false alarms and 0.947 task accuracy, with mean delay remaining between 5.23 and 5.90 slots. By contrast, lowering the belief gate to 0.995 reduces delay to 3.64 slots but introduces a 3.3 percent false alarm rate. This pattern indicates that the passive gates mainly control when confirmation is requested, while the learned belief gate controls the final reliability and delay balance after the full evidence vector has been formed.

Table 13. Sensitivity of CoSMA DAI to the principal evidence and belief parameters.

Configuration	Detection Rate	Delay	Accuracy	False Alarm Rate	Overhead, kbit
persistent gate = 7.5	100.0%	5.62	0.947	0.0%	16.81
persistent gate = 9.0	100.0%	5.42	0.947	0.0%	16.65
persistent gate = 11.0	100.0%	5.51	0.947	0.0%	16.65
persistent gate = 13.0	100.0%	5.81	0.947	0.0%	16.65
local gate = 12.0	100.0%	5.90	0.947	0.0%	16.91
local gate = 15.0	100.0%	5.62	0.947	0.0%	16.81
local gate = 18.0	100.0%	5.23	0.947	0.0%	16.75
ML belief gate = 0.9950	100.0%	3.64	0.947	3.3%	16.61
ML belief gate = 0.9990	100.0%	5.62	0.947	0.0%	16.81
ML belief gate = 0.9995	100.0%	6.86	0.947	0.0%	16.94
persistence = 1	100.0%	8.48	0.947	0.0%	18.09
persistence = 2	100.0%	5.96	0.947	0.0%	17.01
persistence = 3	100.0%	5.62	0.947	0.0%	16.81

Notes: Bold numeric values identify the best result in each column, including ties. Higher detection rate and accuracy are preferred; lower delay, false alarm rate and overhead are preferred.

Dedicated stress scenarios evaluate the separate history-dependent policy described in Section 4.5. Robust fallback is activated when fused evidence exceeds its gate during an unsafe-channel interval. Predictive resynchronisation is activated at a channel-safe candidate more than 30 slots after a preceding repair, without a fresh protected probe. Their activation counts are provided in Table 14 and Figure 12.

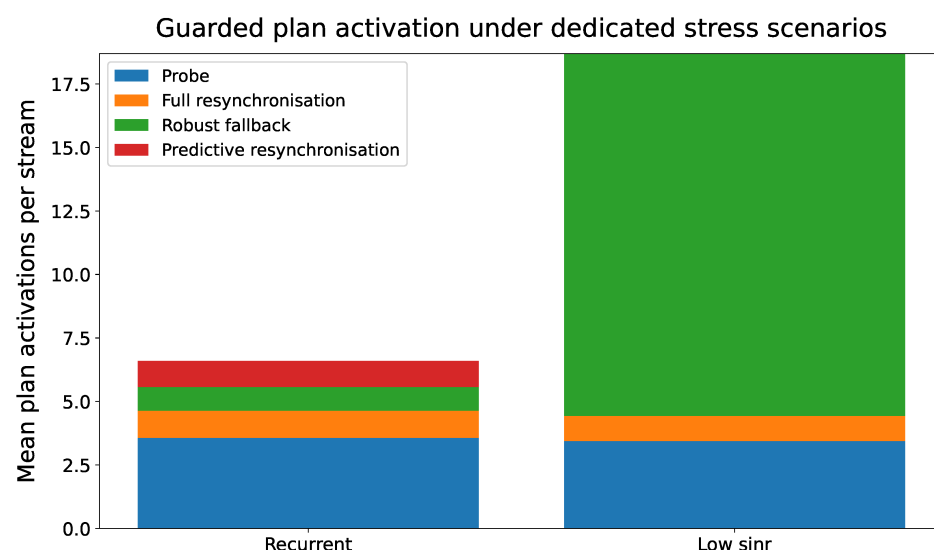


Figure 12. Mean activation counts under the dedicated stress scenarios.

The recurrent scenario produces 3.57 probes, 2.10 repairs and 1.03 predictive repairs per stream on average. The low signal-to-interference-plus-noise ratio scenario produces 14.27 robust fallback activations and no predictive repair, which confirms that the unsafe

channel guard can dominate a high-mismatch candidate. These outcomes verify that the less frequent plans are executable and that the plan library does not collapse into a single probe-and-repair path under all stress conditions.

Table 14. Mean guarded plan activations over 30 stress streams.

Stress Scenario	Probes	Repairs	Robust Fallbacks	Predictive Repairs	Overhead, kbit
Recurrent	3.57	2.10	0.93	1.03	19.30
Low sinr	3.43	1.00	14.27	0.00	10.08

5.8. Separability, Operating Boundary and Learned Belief Validation

The learned belief is validated at the slot level before its use inside the protocol is interpreted. Table 15 reports the class-conditional distributions, the areas under the receiver operating characteristic and precision-recall curves and an explicit histogram overlap between matched and mismatched slots and Figure 13 shows the corresponding curves. The belief reaches an area under the receiver operating characteristic curve of 0.99997 and an average precision of 0.99993, against 0.9478 for the single conditioned scalar that is supplied to the sequential monitors. The class overlap is 0.00039 for the belief and 0.0814 for the scalar, so the two classes are close to separable in the learned representation but are not separable in the scalar. At the selected gate of 0.999, 0.0012 percent of matched slots and 98.97 percent of mismatched slots exceed the threshold, which explains both the zero observed false repair rate and the small delay penalty relative to the fastest monitors.

Table 15. Slot-level separability of the learned mismatch belief against the single conditioned scalar supplied to the sequential monitors. The belief gate is 0.999.

Quantity	Value	Quantity	Value
Matched slots evaluated	85 271	Mismatched slots evaluated	19 729
Belief AUROC	0.99997	Conditioned scalar AUROC	0.94781
Belief average precision	0.99993	Conditioned scalar average precision	0.94030
Belief histogram overlap	0.00039	Conditioned scalar histogram overlap	0.08142
Median belief, matched	1.52×10^{-4}	Median belief, mismatched	0.99985
99.9th percentile, matched	0.0336	0.1st percentile, mismatched	0.7388
Matched slots above the gate	0.0012%	Mismatched slots above the gate	98.97%

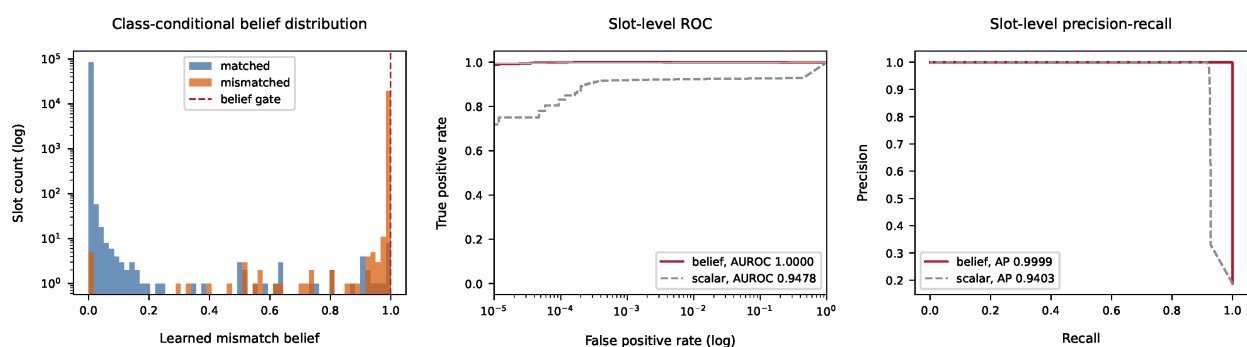


Figure 13. Class-conditional distribution of the learned mismatch belief, slot-level receiver operating characteristic and precision-recall curves, compared against the single-conditioned scalar supplied to the sequential monitors.

The reported separability is a property of the observation model rather than of the estimator alone. A codebook displacement changes the transmitted fingerprint directly, so the evidence is close to deterministic once the displacement exceeds the channel noise floor. This is why the sequential monitors also reach full detection on the principal scenarios and why the parity study of Section 5.6 finds the estimator interchangeable. The correct

interpretation of the perfect event-level scores is therefore that the benchmark is separable at the evaluated mismatch magnitudes, not that the estimator is uniformly superior.

Table 16 and Figure 14 characterise the boundary directly by fixing the mismatch magnitude α and reducing it toward zero. Two regimes are visible. Above $\alpha = 0.30$ the divergence causes substantial functional harm and every method detects it, with the monitors faster than the proposed protocol because no probe confirmation is required. Below $\alpha = 0.10$ the functional outage falls to the residual error floor of the link, so there is almost nothing to repair, yet the proposed protocol still detects the displacement at 100 percent and commits a full resynchronisation. The sequential monitors do not and Reference CTM does not detect at all below $\alpha = 0.10$.

The consequence for the operating point, namely that the evidence process responds to representational displacement rather than to functional harm, is discussed in Section 5.13.

Table 16. Behaviour as the mismatch magnitude α approaches the practical detection boundary. Each cell reports the detection rate in percent and the mean delay in slots. The second column is the fraction of accepted post-onset slots that are genuinely functionally mismatched at that magnitude.

α	True Outage	CoSMA DAI	Page Hinkley	Reference CTM
0.02	0.0037	100.0/2.33	0.0/N/A	0.0/N/A
0.04	0.0037	100.0/2.33	0.0/N/A	0.0/N/A
0.06	0.0037	100.0/2.33	66.7/71.20	0.0/N/A
0.08	0.0037	100.0/2.33	100.0/24.30	0.0/N/A
0.10	0.0037	100.0/2.33	100.0/14.10	46.7/114.29
0.15	0.0093	100.0/2.33	100.0/7.10	100.0/20.30
0.20	0.0112	100.0/2.33	100.0/3.27	100.0/11.13
0.30	0.0478	100.0/2.33	100.0/1.50	100.0/5.17
0.50	0.3887	100.0/2.33	100.0/0.57	100.0/2.27
0.75	0.9959	100.0/2.33	100.0/0.37	100.0/1.73
1.00	1.0000	100.0/2.33	100.0/0.30	100.0/1.53

N/A indicates that the mean delay is undefined because no event was detected at that mismatch magnitude.

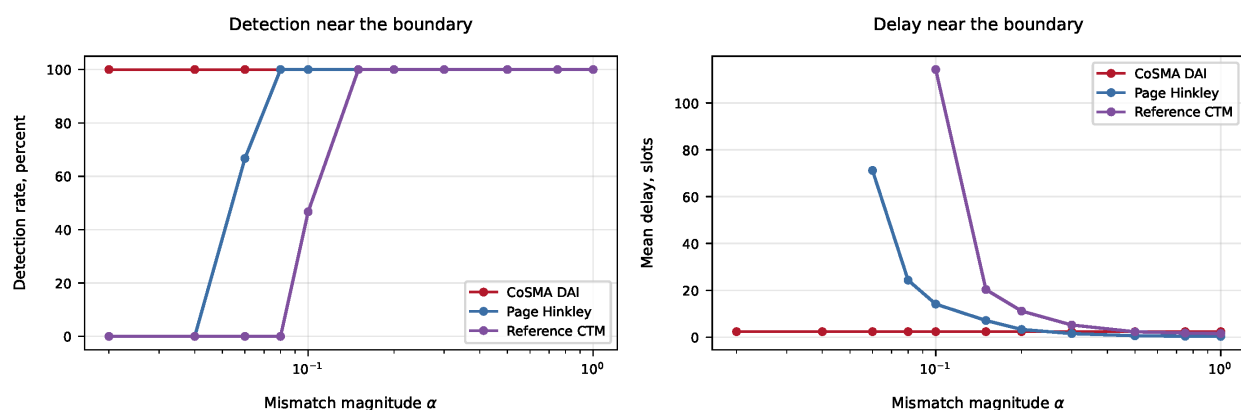


Figure 14. Detection rate and delay as the mismatch magnitude is reduced toward the practical detection boundary.

5.9. Zero Shot Generalisation to Unseen Mismatch and Unseen Channels

The frozen belief model and the frozen configuration are applied without retraining and without threshold reselection to conditions that were never used for model fitting, gate selection or comparator selection. Table 17 reports 7 unseen mismatch families: codebook merging, merging with a gradual onset, pruning from 128 to 96 entries, independent re-indexing of a codeword subset, targeted divergence restricted to confusable class pairs, uniformly random re-assignment of thirty percent of entries and receiver-side decoder

drift. Table 18 reports 5 unseen link models, each accompanied by a repeated matched null campaign of 150 streams so that false repair on an unseen channel is measured rather than assumed. Table 19 reports transfer to codebooks of 256 and 512 entries with wider latent spaces. Figure 15 summarises both campaigns.

Table 17. Zero-shot transfer to mismatch families that were never used for belief training, gate selection or comparator selection. The frozen model and frozen configuration are applied without any retraining.

Mismatch Family	Group	Det. %	Delay	Outage Rate	Recon. dB
abrupt	trained	100.0	2.33	0.0118	21.63
gradual	trained	100.0	4.97	0.0004	21.85
sparse	trained	100.0	9.57	0.0096	21.73
merge	unseen	100.0	2.27	0.0025	21.81
merge gradual	unseen	100.0	9.63	0.0020	21.85
prune	unseen	100.0	14.67	0.0043	21.78
permute	unseen	100.0	12.83	0.0102	21.56
confusable	unseen	100.0	5.97	0.0122	21.63
random subset	unseen	100.0	10.97	0.0132	21.52
decoder drift	unseen	0.0	N/A	0.1927	16.78

N/A indicates that mean detection delay is undefined because no event was detected.

Table 18. Zero-shot channel transfer. The belief model is trained only on the Rician $K = 4$ link. Matched null campaigns of 150 streams are repeated on every link so that false repair under an unseen channel is measured rather than assumed.

Channel	Method	Det. %	FAR %	FAR Upper %	Delay	Outage Rate
rician k4	CoSMA DAI	100.0	0.00	2.43	5.62	0.0073
rician k4	Page Hinkley	100.0	0.67	3.66	3.39	0.0034
rician k4	Reference CTM	100.0	0.00	2.43	11.58	0.0120
rician k1	CoSMA DAI	100.0	0.00	2.43	5.01	0.0067
rician k1	Page Hinkley	96.7	0.67	3.66	3.95	0.0294
rician k1	Reference CTM	100.0	0.00	2.43	13.90	0.0151
rayleigh	CoSMA DAI	100.0	0.00	2.43	6.09	0.0078
rayleigh	Page Hinkley	100.0	0.00	2.43	3.92	0.0037
rayleigh	Reference CTM	100.0	0.00	2.43	17.42	0.0177
awgn	CoSMA DAI	100.0	0.00	2.43	5.60	0.0074
awgn	Page Hinkley	100.0	0.00	2.43	2.56	0.0026
awgn	Reference CTM	100.0	0.00	2.43	9.96	0.0092
high doppler	CoSMA DAI	100.0	0.00	2.43	5.94	0.0077
high doppler	Page Hinkley	96.7	0.00	2.43	3.76	0.0288
high doppler	Reference CTM	100.0	0.00	2.43	15.87	0.0161
bursty	CoSMA DAI	100.0	0.00	2.43	7.04	0.0080
bursty	Page Hinkley	100.0	0.00	2.43	4.64	0.0040
bursty	Reference CTM	100.0	0.00	2.43	17.51	0.0164

Table 19. Zero-shot transfer to larger semantic codebooks. The frozen belief model and frozen gates are reused without retraining while the codebook grows from 128 to 512 entries.

N_c	Ceiling	Method	Det. %	FAR %	Delay	Outage Rate	Recon. dB
128	0.919	CoSMA DAI	100.0	0.00	6.30	0.0081	21.72
128	0.919	Page Hinkley	100.0	1.00	3.20	0.0032	21.80
128	0.919	Reference CTM	100.0	0.00	10.77	0.0110	21.70
128	0.919	Task label only	0.0	0.00	N/A	0.7366	15.96
256	0.933	CoSMA DAI	100.0	1.00	7.12	0.0080	22.25
256	0.933	Page Hinkley	100.0	1.00	3.28	0.0028	22.34
256	0.933	Reference CTM	100.0	0.00	10.23	0.0093	22.24
256	0.933	Task label only	0.0	0.00	N/A	0.7296	16.48

Table 19. Cont.

N_c	Ceiling	Method	Det. %	FAR %	Delay	Outage Rate	Recon. dB
512	0.956	CoSMA DAI	100.0	0.00	6.40	0.0077	20.66
512	0.956	Page Hinkley	100.0	1.00	3.03	0.0029	20.73
512	0.956	Reference CTM	100.0	0.00	9.50	0.0099	20.64
512	0.956	Task label only	0.0	0.00	N/A	0.7382	14.86

N/A indicates that detection delay is not applicable to the task-label-only baseline, which performs no mismatch detection or repair.

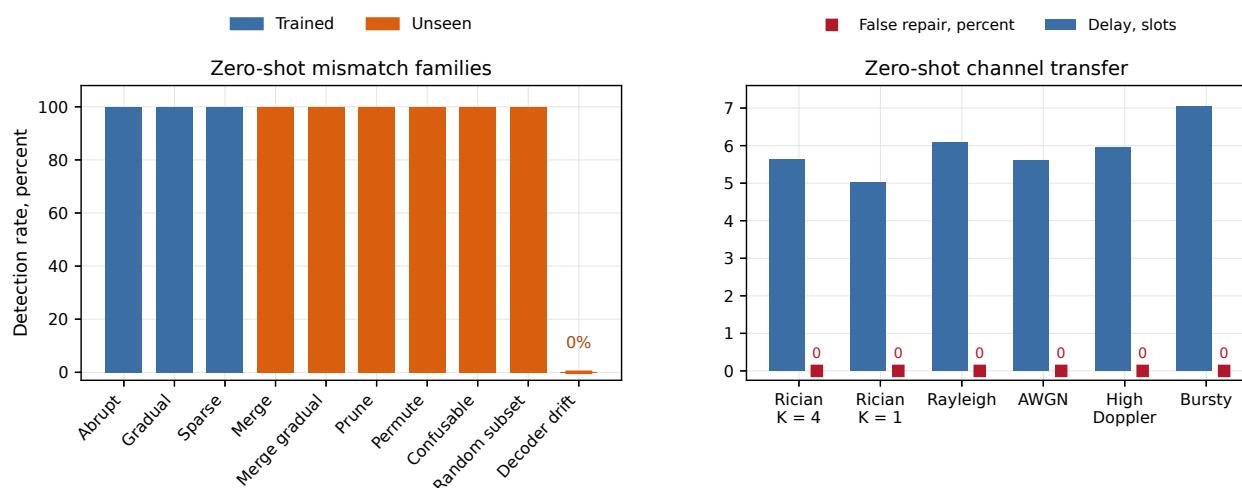


Figure 15. Zero-shot transfer, showing detection over trained and unseen mismatch families on the left and delay and false repair over link models on the right, where red squares mark zero observed false repair rates.

Detection is retained at 100 percent with zero observed false repairs on every link and in 6 of the 7 unseen mismatch families, at delays between 2.27 and 10.57 slots. The comparators do not transfer as consistently. Page Hinkley loses detection on the Rician $K = 1$ and high-Doppler links, where its residual outage rate rises from 0.0034 to approximately 0.029 and the Diagnostic E Process retains a nonzero false repair rate on three links. The proposed protocol is therefore the only evaluated method that retains full detection with zero observed false repair across all six link models. Transfer to the larger codebooks preserves full detection, a false repair rate between 0.0 and 1.0 percent and a residual outage rate near 0.008, which indicates that the conditioned evidence representation is not tied to the 128-entry configuration used for training.

The remaining family is a negative result: receiver-side decoder drift leaves the codebook and therefore the transmitted fingerprint, almost unchanged, so a fingerprint-based evidence process is structurally blind to it. Detection falls to zero percent and the residual outage rate reaches 0.1927. Every comparator fails on the same family, which locates the limitation in the observation model rather than in the belief estimator or the guarded plan library. Decoder-side attestation, for example, a periodic protected response computed by the receiver task head and compared against a transmitter reference, is identified as the required extension.

The campaigns above hold the estimator fixed and vary the tested condition. A stronger protocol is also reported in which the estimator itself is refitted so that the tested condition is absent from its training distribution by construction. Table 20 reports a leave-one-mechanism-out evaluation in which the belief is retrained on two of the three divergence mechanisms and evaluated on the held-out third and Table 21 reports cross-channel retraining in which the belief is fitted on one link model and evaluated on link

models with different fading and interference statistics. The gates, the plan library and the protocol remain the frozen principal ones, so only the estimator changes.

Detection remains at 100 percent in every held-out mechanism and in every train-test channel pair. Delay changes in the expected direction, falling to 1.83 slots when the abrupt mechanism is held out and rising to 7.83 slots when the sparse mechanism is held out, which is the hardest geometry and the one whose class-local evidence pattern is not represented in the reduced training set. The refitted models show one false repair in 120 matched null streams, a rate of 0.83 percent, against zero for the principal model. This is attributed to the reduced training budget used for the retraining protocol, which employs 40 rather than 60 drift seeds and 120 rather than 200 null seeds and the budget is not enlarged to remove it. The evaluation is therefore an out-of-distribution test with respect to the divergence mechanism and to the channel statistics, rather than an interpolation across magnitudes of a single generator.

Table 20. Leave-one-mechanism-out evaluation. The belief estimator is refitted without the tested divergence mechanism and the gates and protocol remain the frozen principal ones. The final row is the principal in-distribution model.

Held Out	Trained on	AUROC	Det. %	FAR %	Delay	Outage Rate
abrupt	gradual + sparse	1.00000	100.0	0.83	1.83	0.0101
gradual	abrupt + sparse	1.00000	100.0	0.83	6.50	0.0010
sparse	abrupt + gradual	0.99988	100.0	0.83	7.83	0.0085
none (principal model)	abrupt + gradual + sparse	1.00000	100.0	0.00	5.62	0.0073

Table 21. Cross-channel transfer with retraining. The belief estimator is refitted on one link model and evaluated on link models with different fading and interference statistics. Bold rows are in-distribution.

Trained on	Tested on	Det. %	FAR %	FAR Upper %	Delay	Outage Rate
rician k4	rician k4	100.0	0.83	4.56	4.38	0.0049
rician k4	rayleigh	100.0	0.83	4.56	5.19	0.0056
rician k4	awgn	100.0	0.83	4.56	4.53	0.0050
rician k4	high doppler	100.0	0.83	4.56	4.84	0.0052
rician k4	bursty	100.0	0.83	4.56	6.38	0.0061
rayleigh	rician k4	100.0	0.83	4.56	4.98	0.0060
rayleigh	rayleigh	100.0	0.83	4.56	5.37	0.0062
rayleigh	awgn	100.0	0.83	4.56	4.90	0.0059
rayleigh	high doppler	100.0	0.83	4.56	5.23	0.0060
rayleigh	bursty	100.0	0.83	4.56	6.51	0.0068
awgn	rician k4	100.0	0.83	4.56	4.88	0.0058
awgn	rayleigh	100.0	0.83	4.56	5.54	0.0064
awgn	awgn	100.0	0.83	4.56	4.98	0.0059
awgn	high doppler	100.0	0.83	4.56	5.29	0.0062
awgn	bursty	100.0	0.83	4.56	6.71	0.0068

Transfer to a previously unseen dataset is also reported. Table 22 applies the frozen belief estimator, fitted on the 8 by 8 digit source, to two independent 28-by-28 sources, MNIST and Fashion-MNIST. Each source receives its own learned projection, codebook, task classifier and channel-conditioning tables, exactly as the principal pipeline constructs them, so only the belief estimator and the gates are carried across. Detection remains at 100 percent on both sources across the three trained and four unseen mismatch families, with a matched null false repair rate of 0.67 percent on MNIST and zero on Fashion-MNIST and mean delays of 5.31 and 6.84 slots against 6.57 slots on the training source. The evaluation therefore spans three sources.

Table 22. Transfer to previously unseen datasets. The belief estimator was fitted on the 8×8 digit source and is applied without retraining to two independent 28×28 sources, each with its own learned projection, codebook, task classifier and conditioning tables. Results are averaged over the three trained and four unseen mismatch families.

Source	Ceiling	Det. %	FAR %	FAR Upper %	Delay	Outage Rate
digits 8×8 (training source)	0.9194	100.0	0.00	2.43	6.57	0.0074
MNIST 28×28	0.8215	100.0	0.67	3.66	5.31	0.0070
Fashion-MNIST 28×28	0.7470	100.0	0.00	2.43	6.84	0.0093

Transfer to natural-image sources is reported in Table 23 and scaling of the latent dimension and codebook on those sources in Table 24. CIFAR-10 and CIFAR-100 are encoded by an ImageNet-pretrained MobileNetV3-small that is not fine-tuned, giving 576-dimensional features from which the principal projection, codebook, task head and conditioning tables are built as for every other source; the belief estimator and gates remain frozen. On CIFAR-10 detection is 100 percent on three of four links and 99.4 percent under AWGN, with zero observed false repairs in 300 null streams per link and delays between 3.30 and 7.91 slots. CIFAR-100 is harder: detection is 98.9 to 100 percent and the delay rises to between 15.29 and 16.85 slots, because the hundred-class task head yields a codebook whose entries are closer together in fingerprint space and the divergence takes longer to exceed the channel noise floor. Growing the representation to a 128-dimensional latent and a 2048-entry codebook on CIFAR-100 lowers detection to 98.0 percent and raises the delay to between 22.10 and 24.88 slots, while the false repair rate stays at zero. These failures at scale are reported as they occur; the eight-dimensional fingerprint is not enlarged to remove them and its observability limit at large codebooks is stated in Section 5.13. These results were produced with a separate execution environment and are imported unmodified together with their provenance.

Table 23. Frozen-detector transfer to CIFAR-10 and CIFAR-100. Each source is encoded by an ImageNet-pretrained MobileNetV3-small without fine-tuning, followed by the principal projection, codebook, task head and conditioning tables; the belief estimator and gates are unchanged. Detection is over the three trained families and the unseen re-indexing and additive-perturbation families. Codebook NMSE is the residual normalised squared disagreement between receiver and transmitter codewords after repair.

Source	Channel	Families	Det. %	FAR %	Null Streams	Delay	Codebook NMSE
digits 8×8	rician	6	100.0	0.33	300	4.83	0.0110
digits 8×8	awgn	6	97.8	0.00	300	4.01	0.0261
digits 8×8	rayleigh	6	100.0	0.67	300	6.32	0.0144
digits 8×8	bursty	6	100.0	0.00	300	7.59	0.0179
CIFAR-10	rician	6	100.0	0.00	300	4.99	0.0166
CIFAR-10	awgn	6	99.4	0.00	300	3.30	0.0188
CIFAR-10	rayleigh	6	100.0	0.00	300	6.98	0.0236
CIFAR-10	bursty	6	100.0	0.00	300	7.91	0.0285
CIFAR-100	rician	6	98.9	0.00	300	15.29	0.0243
CIFAR-100	awgn	6	98.9	0.00	300	16.85	0.0141
CIFAR-100	rayleigh	6	98.9	0.00	300	15.88	0.0247
CIFAR-100	bursty	6	100.0	0.00	300	16.08	0.0317

Table 24. Latent dimension and codebook size on CIFAR-100 features with the frozen belief estimator and frozen gates. Detection is over five mismatch families; 300 matched null streams per row.

d_z	N_c	Channel	Det. %	FAR %	Delay
24	128	rayleigh	99.3	0.00	23.04
24	128	rician	98.7	0.00	18.40
96	1024	rayleigh	99.3	0.00	21.85

Table 24. *Cont.*

d_z	N_c	Channel	Det. %	FAR %	Delay
96	1024	rician	100.0	0.00	17.82
128	2048	rayleigh	98.0	0.00	24.88
128	2048	rician	98.0	0.00	22.10

5.10. Protocol-Level Fairness, Belief Calibration, Paired Statistics and Auxiliary Cost

This section addresses five questions that the principal comparison leaves open, namely whether the development budgets are equal, what the guard layer contributes beyond decision structuring, whether the learned score can be read as a probability, whether the reported differences are statistically supported on the paired streams and what the auxiliary traffic costs are once it is normalised by useful output and it closes with the authentication of the task belief.

The first question concerns the development budget. The principal comparison tunes each comparator on 10 drift and 30 matched null development streams, while the learned belief is fitted on 60 drift and 200 null development streams, so the candidate counts are equal, but the development data are not. Table 25 repeats the selection for every comparator on the same enlarged pool with the same 144 candidates. The comparators improve materially. CUSUM rises from 90.00 to 100.00 percent detection and its false repair rate falls from 4.67 to 0.67 percent, the Diagnostic E Process reaches a zero observed false repair rate and Conditional CTM reaches 100 percent detection with a zero observed false repair rate at 3.76 slots, against 5.62 slots for the proposed protocol. The delay advantage of Section 5.4 over the zero false repair comparator therefore holds against Reference CTM, but not against a comparator that has been given the same development data and it is reported as comparator-specific.

Table 25. Effect of equalising the development budget. Under the principal budget each comparator is tuned on 10 drift and 30 matched null development streams; under the equal budget, every comparator is reselected on the same 60 drift and 200 matched null development streams that the learned belief is fitted on, with the same 144 candidates. Held-out evaluation is identical in both cases.

Budget	Method	Det. %	FAR %	Delay	Outage Rate
principal budget	CUSUM	90.0	4.67	2.04	0.0774
principal budget	Page Hinkley	100.0	0.67	3.39	0.0034
principal budget	WATCH	100.0	0.33	3.84	0.0038
principal budget	Diagnostic E Process	100.0	2.00	7.87	0.0091
principal budget	Conditional CTM	100.0	0.33	3.60	0.0034
principal budget	Reference CTM	100.0	0.00	11.58	0.0120
principal budget	CoSMA DAI	100.0	0.00	5.62	0.0073
equal budget	CUSUM	100.0	0.67	3.30	0.0033
equal budget	Page Hinkley	100.0	0.67	3.27	0.0033
equal budget	WATCH	100.0	0.33	3.74	0.0038
equal budget	Diagnostic E Process	100.0	0.00	9.94	0.0118
equal budget	Conditional CTM	100.0	0.00	3.76	0.0037
equal budget	Reference CTM	100.0	0.00	11.58	0.0120
equal budget	CoSMA DAI	100.0	0.00	5.62	0.0073

The second question is what the guard layer contributes, given that the equivalence between the plan library and the independently written threshold cascade of Section 5.7 holds only when the guards are present in both. Table 26 removes the channel safety guard and imposes a deep fade over the divergence interval. Detection is unchanged at 100 percent and the unguarded variant is 26.63 slots faster, because it commits as

soon as the belief crosses the gate. It also commits 85.56 percent of its repairs below the safety threshold, at a median signal-to-interference-plus-noise ratio of -2.16 dB, where the expected transmitted bits per useful bit of the codebook transfer rise to 9.95, against 1.06 for the guarded protocol committing at a median of 3.20 dB. The quantitative benefit of the guard layer is therefore a reduction of roughly an order of magnitude in the expected cost of the transfer, purchased with additional delay and this trade is the measurable content of the BDIx layer. The deliberation formalism itself supplies auditability rather than discrimination and no detection benefit is claimed for it.

Table 26. Quantified contribution of the guard layer under a deep fade imposed over the divergence interval. The retransmission multiplier is the expected transmitted bits per useful bit of the codebook transfer, obtained from the block error probability at the committing signal-to-interference-plus-noise ratio.

Configuration	Det. %	Delay	Commits Below Guard %	Median Commit SINR dB	Median Multiplier	Mean Multiplier
guarded plan library	100.0	33.56	0.00	3.20	1.06	1.08
guards removed	100.0	6.93	85.56	-2.16	9.95	12,823.33

The third question is whether the learned score B_t can be read as a probability, which Table 27 and Figure 16 address by measuring its calibration. The expected calibration error is 0.00037 and the Brier skill score is 0.9980 against the base rate, which is strong in aggregate and at the selected gate the mean reported value of 0.99983 agrees with the empirical frequency of the imposed-divergence target of 0.99995. The maximum calibration error is nevertheless 0.704, concentrated in sparsely populated intermediate bins where the estimator is rarely evaluated because the two classes are close to separable. Isotonic recalibration on a disjoint development split reduces the expected calibration error but does not remove the intermediate-bin error and does not change any protocol decision, since the gate lies far above that region. The quantity is therefore reported as a learned mismatch score with measured calibration, not as a calibrated posterior.

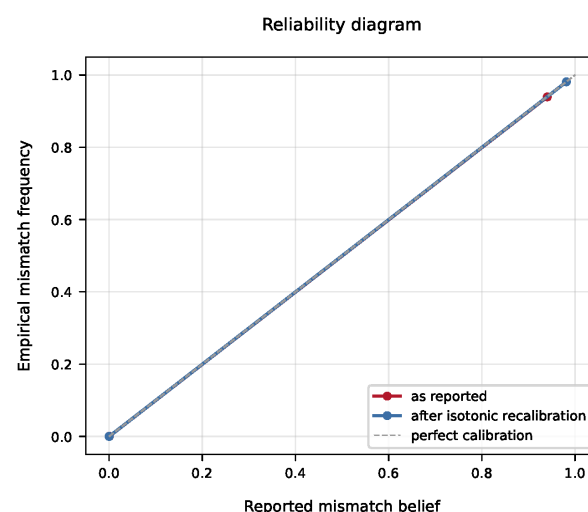


Figure 16. Reliability diagram of the learned mismatch score, before and after isotonic recalibration on a disjoint development split.

The fourth question is whether the reported differences are statistically supported. All methods are evaluated on identical realisations, so Table 28 reports paired differences with 2000-sample bootstrap intervals and two-sided Wilcoxon signed-rank tests over the 90 shared held-out streams. The delay reduction against Reference CTM is 5.96 slots with a 95 percent interval of 2.70 to 9.16 slots and $p = 7 \times 10^{-6}$, so that comparison is supported.

The proposed protocol is significantly slower than Page Hinkley by 2.23 slots and the differences against WATCH and Conditional CTM are not statistically significant at the 5 percent level. Residual outage differences are small and are significant in both directions, depending on the comparator. The delay claim in the abstract and conclusion is restricted to the paired comparisons that survive this test.

Table 27. Probability calibration of the learned mismatch belief, measured over the matched null and held-out drift streams. Isotonic recalibration is fitted on a disjoint development split.

Quantity	Value	Quantity	Value
Positive base rate	0.1879	Brier score	0.000302
Expected calibration error	0.00037	Brier skill score	0.99802
Maximum calibration error	0.7037	Expected calibration error after isotonic	0.00030
Nominal belief above the gate	0.99983	Empirical frequency above the gate	0.99995

Table 28. Paired comparison over the 90 shared held-out streams. Differences are CoSMA DAI minus the comparator, so a negative delay difference favours the proposed method. Intervals are 2000-sample paired bootstrap intervals and p values are two-sided Wilcoxon signed-rank tests.

Comparator	Δ Delay	95% CI	p	Δ Outage	p
CUSUM	3.81	[2.35, 5.52]	2.9×10^{-10}	−0.07015	2.8×10^{-2}
Page Hinkley	2.23	[0.86, 3.90]	1.8×10^{-2}	0.00387	2.9×10^{-3}
WATCH	1.78	[0.37, 3.50]	1.4×10^{-1}	0.00350	1.2×10^{-2}
Diagnostic E Process	−2.24	[−3.79, −0.52]	4.8×10^{-6}	−0.00180	5.1×10^{-3}
Conditional CTM	2.02	[0.60, 3.74]	6.9×10^{-2}	0.00383	3.0×10^{-3}
Reference CTM	−5.96	[−9.16, −2.70]	7.0×10^{-6}	−0.00476	1.0×10^{-3}
CUSUM + task belief	3.81	[2.35, 5.52]	2.9×10^{-10}	−0.07015	2.8×10^{-2}
Page Hinkley + task belief	2.23	[0.86, 3.90]	1.8×10^{-2}	0.00387	2.9×10^{-3}
WATCH + task belief	1.78	[0.37, 3.50]	1.4×10^{-1}	0.00350	1.2×10^{-2}
Diagnostic E Process + task belief	−2.24	[−3.79, −0.52]	4.8×10^{-6}	−0.00180	5.1×10^{-3}
Conditional CTM + task belief	2.02	[0.60, 3.74]	6.9×10^{-2}	0.00383	3.0×10^{-3}
Reference CTM + task belief	−5.96	[−9.16, −2.70]	7.0×10^{-6}	−0.00476	1.0×10^{-3}

Delay comparisons use streams detected by both methods: 81 streams for CUSUM and CUSUM with the task belief and 90 for the other monitors; outage comparisons use all 90 shared streams. Values are two-sided Wilcoxon signed-rank p -values with the zero-split convention. The two CUSUM delay p -values were recomputed from the saved paired observations after removal of fixed-decimal rounding.

The fifth question concerns the cost of the auxiliary traffic once it is normalised by useful output, because the control-share reduction reported in Section 5.2 depends on classifying task-belief bits as semantic side information rather than as control traffic. Table 29 removes that dependence by counting every non-payload bit as auxiliary and normalising by the number of correct downstream decisions. Under this accounting, the proposed protocol requires 61.57 auxiliary bits and 1.4396 μ J per correct decision, which is higher than every scalar monitor and higher than the label-augmented monitors and is exceeded only by CUSUM, whose missed detections reduce the number of correct decisions. The proposed protocol therefore does not reduce auxiliary traffic or energy per useful decision and its advantage lies in reliability-constrained detection and in the residual semantic outage it removes.

A final point concerns authentication, since the protocol assumes an authenticated task belief and the mechanism and its cost should be stated. The construction is a shared symmetric key established during resynchronisation and a truncated message authentication code over the belief field and a monotone sequence counter, which prevents replay and injection of the auxiliary decision. Table 30 charges this explicitly. A 32-bit tag applied to every packet is not viable, since it reaches 55.95 percent of all transmitted bits, whereas a 64-bit tag computed over an epoch of 100 packets costs 2.48 percent and bounds the exposure to one epoch. The epoch construction is adopted and its cost is included in the auxiliary accounting of Table 29.

Table 29. Total auxiliary traffic and radio energy normalised per correct downstream task decision. Auxiliary traffic is every non-payload bit, so the task-belief header, the protected probes and the codebook transfer are counted together and the classification of task-belief bits as side information no longer affects the ordering.

Method	Auxiliary Bits per Correct Decision	Radio Energy μJ per Correct Decision
Task label only	27.55	1.0995
Page Hinkley	34.31	1.2709
Conditional CTM	34.31	1.2709
WATCH	34.32	1.2712
Diagnostic E Process	34.50	1.2782
Reference CTM	34.58	1.2809
CUSUM + task belief	58.02	1.4041
Page Hinkley + task belief	58.02	1.4041
WATCH + task belief	58.02	1.4041
Diagnostic E Process + task belief	58.02	1.4041
Conditional CTM + task belief	58.02	1.4041
Reference CTM + task belief	58.02	1.4041
CoSMA DAI	61.57	1.4396
CUSUM	278.17	10.3042

Table 30. Authentication accounting for the task-belief header. An authenticated belief is assumed throughout and the tag is charged. A truncated message authentication code applied per packet dominates the auxiliary budget, whereas a tag computed over an epoch of packets is affordable.

Scheme	Auth Bits per Stream	Auxiliary kbit	Share of Auxiliary %	Share of Total %
none	0	15.29	0.00	0.00
truncated MAC, 16 bit per packet	24,000	39.29	61.08	38.84
truncated MAC, 32 bit per packet	48,000	63.29	75.84	55.95
MAC over an epoch of 50 packets, 64 bit	1920	17.21	11.16	4.84
MAC over an epoch of 100 packets, 64 bit	960	16.25	5.91	2.48

5.11. Representation Scale, Task Belief Scaling and Calibration Burden

Three scaling questions are addressed with the frozen belief model and the frozen gates. Table 31 reports zero-shot behaviour as the latent dimension grows from 24 to 64 and the codebook grows from 128 to 2048 entries. The source is augmented by small shifts and additive jitter so that a codebook larger than the original training set can be learned without degenerate clusters, which lowers the quantised ceiling relative to the unaugmented benchmark and therefore makes the task harder rather than easier. Full detection is retained at every configuration, the matched null false repair rate remains at or below 0.83 percent and the delay remains between 4.90 and 5.83 slots. The reported reliability is therefore not an artefact of the 128-entry, 24-dimensional configuration used for training.

The task-preservation belief scales far less favourably. Table 32 measures the coded header cost on compound label spaces of 10 to 160 classes constructed from the same source and Table 33 extends the accounting analytically to structured outputs. For a ten-class task, the coded header occupies 12 bits against 45 bits of coded semantic payload, a ratio of 0.27, which is the regime in which the principal configuration operates. The ratio grows as the logarithm of the alphabet for closed-set classification and reaches 0.67 even at one thousand classes, so top-1 preservation remains viable there. It becomes untenable for structured outputs: twenty detection boxes require sixteen times the payload, thirty-two generated text tokens require thirty-two times and a 64 by 64 segmentation mask requires more than one thousand times. For these tasks, a top-1 belief is not compact common knowledge and the mechanism must be replaced by a compressed or partial posterior, by preservation of a low-rate summary statistic rather than the full decision, or by accepting the semantic

outage during confirmation. The scope of the task-preservation contribution is therefore restricted to closed-set task-oriented links with small output alphabets.

Table 31. Zero-shot behaviour on higher-dimensional semantic representations and substantially larger codebooks. The frozen belief model and frozen gates are reused without retraining. The source is augmented so that large codebooks remain non-degenerate, which lowers the quantised ceiling relative to the unaugmented benchmark.

d_z	N_c	Index Bits	Ceiling	Det. %	FAR %	Delay	Outage Rate
24	128	7	0.670	100.0	0.00	4.90	0.0071
32	256	8	0.752	100.0	0.00	5.65	0.0088
48	512	9	0.795	100.0	0.83	5.32	0.0079
64	1024	10	0.829	100.0	0.00	5.83	0.0085
64	2048	11	0.825	100.0	0.00	5.62	0.0087

Table 32. Task-belief overhead against the size of the closed-set output space, measured with the accounting of Table 2. The coded semantic payload occupies 45 bits per packet.

Classes K	Label Bits	Header Bits	Rescue Bits	Header kbit	Header/Payload
10	4	12	40	6.00	0.267
20	5	15	45	7.50	0.333
40	6	18	50	9.00	0.400
80	7	21	55	10.50	0.467
160	8	24	60	12.00	0.533

Table 33. Analytic extension of the task-belief cost to structured output spaces. The side information is the coded top-1 decision under the same repetition factor. The final column states whether the task belief is cheaper than the semantic index it is intended to protect.

Downstream Task	Tokens	Alphabet	Coded Bits	Ratio	Cheaper
10-class classification	1	10	12	0.27	yes
1000-class classification	1	1000	30	0.67	yes
object detection, 20 boxes	20	4096	720	16.00	no
segmentation, 64×64 mask, 20 classes	4096	20	61,440	1365.33	no
text generation, 32 tokens	32	32,000	1440	32.00	no
continuous control, 8 dims at 16 bit	8	65,536	384	8.53	no

The calibration burden of the channel-conditioned reference tables is quantified in Table 34. The implementation populates 768 cells, six signal-to-interference-plus-noise ratio strata by 128 codewords, with 80 Monte-Carlo samples each and stores four tables occupying 24.0 kB at double precision. Reducing the budget to 40, 20 or 10 samples per cell leaves detection, delay and residual outage essentially unchanged, whereas 5 samples per cell raises the false repair rate to 2.50 percent and 2 samples per cell raises it to 55.00 percent, which locates the practical floor at approximately 10 samples per cell.

The final row of Table 34 also reports a variant in which the codeword axis is removed and only the signal-to-interference-plus-noise ratio stratum is retained. This reduces the table from 768 cells to 6 cells and leaves detection at 100 percent, the false repair rate at zero and the residual outage rate unchanged, so the calibration cost of the conditioning mechanism does not grow with codebook size. The implication for the description of the reference tables is discussed in Section 5.13.

Figure 17 relates the coded task-belief cost to output-alphabet size and shows how a smaller reference-calibration budget affects detection and false repair.

Table 34. Calibration burden of the channel-conditioned reference tables. The final row removes the codeword axis entirely and retains only the SINR stratum, which reduces the table from 768 cells to 6 without any loss of detection or reliability.

Configuration	Cells	Samples/Cell	Total	Memory kB	Det. %	FAR %	Delay
80 samples per cell (principal)	768	80	61,440	24.0	100.0	0.00	6.30
40 samples per cell	768	40	30,720	24.0	100.0	0.83	6.17
20 samples per cell	768	20	15,360	24.0	100.0	0.83	6.33
10 samples per cell	768	10	7680	24.0	100.0	0.83	4.45
5 samples per cell	768	5	3840	24.0	96.7	2.50	5.95
2 samples per cell	768	2	1536	24.0	93.3	55.00	10.41
stratum only, codeword axis removed	6	80	480	0.2	100.0	0.00	6.73

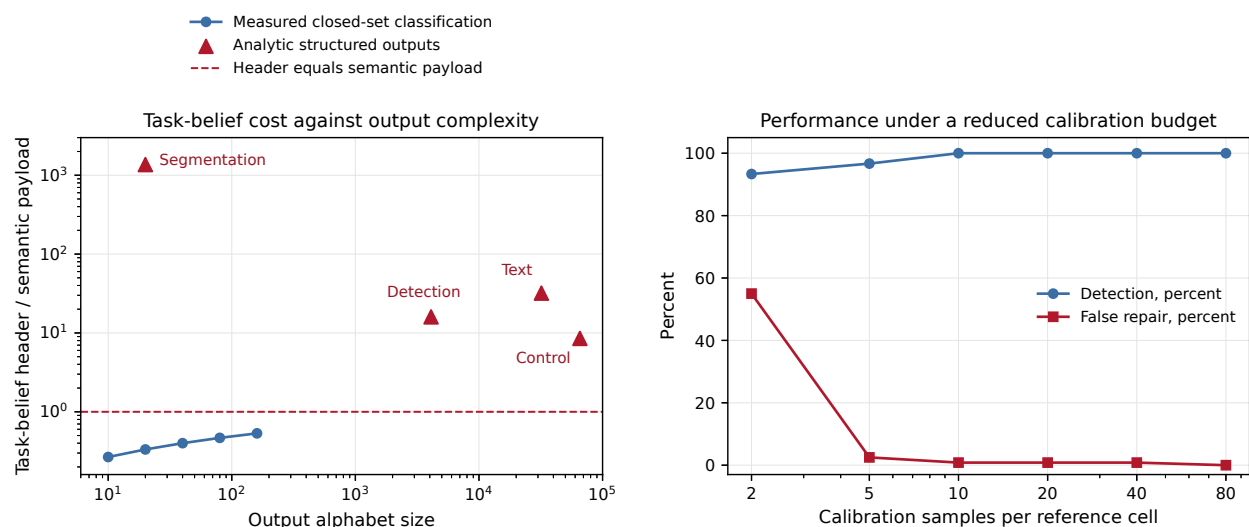


Figure 17. Task-belief header cost relative to coded semantic payload against output-alphabet size on the left, where red triangles represent analytic costs for detection, segmentation, text generation and continuous control and the red dashed line denotes equal cost, with detection and false repair under reduced calibration budgets on the right.

5.12. Deep Learned Representation and Structured Downstream Tasks

The enlarged configurations of Section 5.11 retain the principal-component encoder and the k-means codebook of the principal pipeline. To test the protocol on a learned representation with a vector-quantised bottleneck, a convolutional VQ-VAE is trained on Fashion-MNIST and on MNIST. The encoder maps a 28-by-28 image to a 64-dimensional latent, the latent is quantised against a jointly learned 512-entry codebook with exponential-moving-average updates and dead-code re-seeding and the decoder reconstructs the image from the quantised vector. The single-token bottleneck keeps the transmitted semantic unit identical to the principal pipeline, one codebook index per slot, while replacing both the encoder and the codebook with learned components. The task head is a logistic classifier on the unquantised latent, as before and the belief estimator and gates are the frozen principal ones, so no part of the detection path is retrained.

Three structured downstream tasks are evaluated empirically from the receiver's decoded representation: image reconstruction, foreground segmentation and object localisation. A foreground mask is obtained by thresholding the decoded image and is scored by intersection over the union against the mask of the source image and a bounding box is the extent of that mask and is scored by box intersection over union. None of these can be served by a class decision. The task-label-only control is scored by decoding the class-conditional mean image of the transmitted label, which is the best a four-bit belief

can do for a structured task. Table 35 reports the outcome and Table 36 prices the side information that a task belief would need to carry instead.

Table 35. Deep learned semantic representation with structured downstream tasks. A convolutional VQ-VAE with a 64-dimensional latent and a jointly learned 512-entry codebook is trained on each source; the frozen belief estimator and frozen gates are applied without retraining. Detection is reported over the three trained and four unseen mismatch families. Reconstruction fidelity, foreground-segmentation IoU and bounding-box IoU are computed from the receiver's decoded representation and cannot be served by a class decision.

Method	Det. % Trained/Unseen	FAR %	Delay	Outage Rate	PSNR dB	Seg. IoU	Box IoU
Fashion-MNIST: ceiling 0.771, aligned-path segmentation IoU 0.736, box IoU 0.848							
CoSMA DAI	100.0/100.0	0.00	5.21	0.0063	16.31	0.738	0.847
Page Hinkley	100.0/100.0	0.00	2.68	0.0035	16.35	0.738	0.848
Reference CTM	100.0/100.0	0.00	12.51	0.0129	16.20	0.735	0.844
Oracle repair, no label	100.0/100.0	N/A	0.00	0.0015	16.38	0.739	0.848
Semantic payload, no repair	0.0/0.0	N/A	N/A	0.4874	12.62	0.573	0.712
Task label only	0.0/0.0	N/A	N/A	0.4874	12.62	0.646	0.783
MNIST: ceiling 0.914, aligned-path segmentation IoU 0.636, box IoU 0.853							
CoSMA DAI	100.0/100.0	0.00	7.07	0.0069	15.05	0.639	0.850
Page Hinkley	100.0/100.0	0.67	3.27	0.0035	15.08	0.641	0.851
Reference CTM	100.0/100.0	0.00	16.83	0.0158	15.00	0.637	0.849
Oracle repair, no label	100.0/100.0	N/A	0.00	0.0011	15.10	0.641	0.851
Semantic payload, no repair	0.0/0.0	N/A	N/A	0.4526	12.81	0.494	0.763
Task label only	0.0/0.0	N/A	N/A	0.4526	12.81	0.482	0.739

Notes: N/A means not applicable. The oracle is supplied with the true repair time and is not an alarm detector; the two non-detecting controls provide no detection delay or null-stream alarm test. Group headings give the aligned-path reference values for each dataset.

Table 36. Side information that a task belief would have to carry to serve each structured task on the VQ-VAE link, against the 51-bit coded semantic payload.

Downstream Task	Side-Information Bits	Coded Header Bits	Header/Payload
top-1 class	4	12	0.24
bounding box, 4×5 bit	20	60	1.18
foreground mask, 784 bit	784	2352	46.12
reconstruction, 784×8 bit	6272	18,816	368.94

The frozen protocol transfers to the learned representation. On Fashion-MNIST, detection is 100 percent on both the trained and the unseen mismatch families with a matched null false repair rate of 0.00 percent and a delay of 5.21 slots and on MNIST, the corresponding values are 0.00 percent and 7.07 slots. On the structured tasks, the proposed protocol reaches a segmentation intersection over the union of 0.738 against an aligned-path ceiling of 0.736 and a box intersection over the union of 0.847 against 0.848, so semantic repair restores the structured outputs to within the quantisation limit of the representation. The task-label-only control reaches 0.646 and 0.783 on Fashion-MNIST, above the no-repair control at 0.573 because class-conditional silhouettes of garments carry some shape information, but below every method that repairs the codebook; on MNIST the class-mean digit is a worse mask than the drifted representation itself, 0.482 against 0.494, with repair reaching 0.639. Its reconstruction fidelity of 12.62 dB against 16.31 dB shows that per-instance detail is lost entirely. The class decision that dominated the task-accuracy metric in Section 5.5 is therefore of limited or negative value on a structured task, which is the empirical form of the analytic cost argument in Table 33.

5.13. Discussion, Validity and Limitations

The complete result set shows that the protected probe, the channel-conditioned evidence representation and the guarded plan library provide complementary benefits. The parity study of Section 5.6 shows that the protected probe is the principal reliability improvement, because attaching it to any detector reduces false repair, whereas the specific belief estimator is interchangeable. In the principal first-repair policy, the protected probe remains the control safeguard because the learned score cannot commit repair until active evidence has been obtained under an admissible channel; recurrent predictive repair is evaluated separately using the disclosed history-dependent stress policy. The learned task belief is the principal task preservation improvement because it maintains a source-derived decision while complete semantic repair is pending. Section 5.5 shows that this function maintains 94.73 percent task accuracy in every scenario and is equivalent to direct label transmission, so it is reported as a property of the preservation mechanism rather than as a detection advantage. Channel conditioning and semantic class local evidence remain important because their removal produces substantially slower or less reliable operating points. The fair input comparison prevents calibration quality from being attributed incorrectly to the BDIx architecture.

State-of-the-art methods are compared according to the evidence that can be reproduced fairly. Published semantic communication and alignment methods are compared qualitatively in Table 1 using their actual mechanisms and source-reported outcomes. Quantitative performance is compared only with the six sequential monitors that can be evaluated on identical streams through a common score interface. No invented alignment interval, event cost or latency is assigned to an external semantic communication method.

Internal validity is supported by frozen configurations, disjoint seed groups, identical conditioned inputs and paired held-out streams. Construct validity is supported by separating event reliability, conditional delay, task accuracy, semantic rate, control traffic and radio energy instead of compressing all outcomes into one score. The main internal threat is that development and evaluation streams are generated from the same source domain. This threat is reduced but not removed by the zero-shot campaigns of Section 5.9, in which the mismatch generator, the physical link model and the codebook size are all varied without retraining. External validity remains limited to the evaluated semantic classification and reconstruction tasks and the evaluated mobile channel conditions. Statistical validity is strengthened by 30 held-out seeds and 300 streams in each null campaign, although confidence intervals remain necessary and zero observed false alarms cannot be interpreted as zero underlying probability.

Three limitations follow directly from the evaluation. The task-preservation belief is a closed-set class decision, so on a ten-class benchmark it can satisfy any task-accuracy metric without semantic repair; this is why label-independent outcomes are reported and why a task-label-only control is included in Table 10. The predictive component of the belief is not unique, because a random forest supplied with the identical features and the identical protected probe reaches a shorter delay and a lower residual outage rate. Receiver-side decoder drift is not detectable by the present fingerprint-based observation model for any evaluated method and decoder-side attestation is required before that failure mode can be covered. On CIFAR-100 with a 2048-entry codebook, the eight-dimensional fingerprint no longer separates every divergence from the channel noise floor, so detection falls to 98.0 percent and delay rises above twenty slots; the fingerprint dimension is a design parameter that trades observability against payload and was not enlarged to remove that failure. Three further limitations follow from Sections 5.8 and 5.11. First, the evidence process is sensitive to representational displacement rather than to functional harm. Table 16 shows that below a mismatch magnitude of approximately 0.10, the induced functional

outage has already fallen to the residual error floor of the link, so almost nothing remains to be repaired, yet a candidate is still confirmed and an 8320-bit codebook transfer is still committed. The behaviour is safe, because a completed repair restores alignment, but it is not economical and the operating point reported in the principal comparison therefore reflects displacement sensitivity rather than harm sensitivity. A harm-aware repair guard, in which the learned belief is combined with an estimate of the induced functional outage before the resynchronisation guard is satisfied, is required and is listed among the extensions. Second, the task-preservation belief is economical only for small closed-set output spaces; for detection, segmentation, text generation or continuous control, the coded top-1 decision costs between eight and more than one thousand times the semantic payload, as reported in Table 33, so the mechanism does not transfer to those tasks in its present form. Segmentation and localisation are evaluated empirically on a learned VQ-VAE representation in Section 5.12; text generation and continuous control remain analytic. Third, the channel-conditioned reference tables are described in Section 4.1 as codeword-specific, but in the evaluated link model, the payload consistency score depends on the magnitude of the additive channel noise and not on which codeword was transmitted. The measured across-codeword spread of the reference median exceeds the expected Monte-Carlo scale by a factor between 1.05 and 1.33 across the six strata, which is consistent with sampling noise alone and removing the codeword axis leaves detection, false repair and residual outage unchanged. The codeword axis is therefore redundant as implemented and a codeword-dependent reference would become necessary only if the fingerprint distortion were itself codeword-dependent, for example, under nonlinear amplification or codeword-dependent modulation, which is outside the evaluated link model. Three limitations follow from Section 5.10. The six sequential monitors of the principal comparison are compact implementations of the cited constructions; the full paper-defined algorithms, reported in Table 9, perform worse on this benchmark, so the principal comparison is conservative but is not a comparison against the published algorithms as such. When every comparator is given the same development data, Conditional CTM attains a zero observed false repair rate at 3.76 slots, so the delay advantage at that operating point is not general and is claimed only against Reference CTM, where it is supported by a paired test. The learned belief is a decision score with strong aggregate calibration but a maximum calibration error of 0.704 in sparsely populated intermediate bins, so it should not be consumed as a posterior by any downstream component that operates away from the selected gate. Normalised per correct decision, the protocol requires more auxiliary traffic and more radio energy than every scalar monitor, so no communication-efficiency advantage is claimed.

Several further limitations remain. The digits benchmark is compact and although a reconstruction task and codebooks of 128 to 2048 entries with latent dimensions of 24 to 64 are included, the principal evaluation uses one source domain. The task-preservation belief transmits a compact classifier decision, which is appropriate for a task-oriented classification link but does not preserve a full generative semantic representation. A learned codebook and genuine receiver codeword divergence are used, although text and multi-modal semantic transceivers remain untested and no deployment claim is made; a convolutional VQ-VAE image transceiver is evaluated in Section 5.12. The supervised belief updater is trained on generated mismatch labels from three divergence generators and Section 5.9 establishes zero-shot transfer to seven further mismatch families, five further link models and larger codebooks, although transfer to text and multimodal semantic domains remains untested. Two endpoint agents are represented, while coordination among many simultaneous links remains outside the present scope. The physical model includes fading, interference, coding and cyclic redundancy checking, although measured

radio traces, hardware timing and energy measurements are not included. Thirty held-out seeds and 300 matched null streams provide stronger support than a small seed campaign, although the zero observed false alarm rate still has an upper 95 percent confidence limit of 1.22 percent.

The network-efficiency trade-off reflects the additional semantic side information required for task preservation. The proposed task belief raises the correctly served source equivalent rate and the bandwidth-normalised semantic spectral efficiency because more downstream task decisions are preserved before full semantic repair. Both metrics reach 0.487 under the disclosed units and exceed the scalar monitors by 12.88 percent. These two quantities are determined by the number of correct downstream decisions, so they are equalised once every method receives the same task-belief header, as shown in Section 5.5. The reported advantage over the scalar monitors is therefore attributed to the side information rather than to the detector and the separation between methods is retained in residual semantic outage, control share and absolute control traffic. The task belief is useful semantic side information rather than a control message, so it is excluded from the control numerator and remains included in the total radio resource and energy denominators. The resulting control share is 23.59 percent and is 12.62 percent lower than the monitor value, although the absolute control traffic remains 9.29 kbit because protected probes are transmitted. The control-adjusted resource efficiency of 6.203 source equivalent bit per channel use and the radio energy of 0.393 mJ per stream expose the remaining side information cost. Event-triggered task belief transmission, confidence-adaptive coding or joint rate and belief optimisation could reduce this cost without weakening the protected repair logic.

6. Conclusions and Future Work

A common knowledge-aware semantic alignment framework has been developed through transmitter and receiver BDIX agents. A learned 128-entry semantic codebook, a Rician physical link, coded semantic index delivery and functional codebook divergence have been integrated in one reproducible benchmark. Channel-conditioned global evidence, semantic class local evidence, temporal score history, channel state and protected probe evidence are fused by a machine-learned mismatch belief updater. A second machine-learned task belief is retained from the transmitter representation so that a downstream task decision can be preserved while complete semantic repair is pending. Neither learned belief can bypass the explicit safety, budget and explanation guards.

The held-out evaluation contains 30 drift seeds, 300 matched null streams and 300 degrading channel controls. CoSMA DAI obtains 100.00 percent balanced accuracy, precision, recall, F1 score and Matthews correlation coefficient with zero observed matched null false alarms. Reference CTM reaches the same event classification and reliability values, although its aggregate delay is 11.58 slots compared with 5.62 slots for CoSMA DAI. A paired signed-rank test over the shared streams supports this difference, with a mean reduction of 5.96 slots and a 95 percent interval of 2.70 to 9.16 slots. Once every comparator is retuned on the same development budget, Conditional CTM also reaches a zero observed false repair rate at 3.76 slots, so the delay result is reported as comparator-specific rather than general. The resulting 51.44 percent reduction establishes a comparator-specific delay improvement over Reference CTM at the zero-observed-false-alarm operating point of the principal comparison. The task-preservation belief maintains 94.73 percent task accuracy through every divergence scenario, above the quantised ceiling of the semantic path; because a task-label-only control reaches the same value and every comparator reaches it once the identical header is supplied, this is a property of the mechanism and not a detection advantage and the detectors are separated on residual functional semantic outage, which

falls from 73.69 percent to 0.73 percent over 6.62 slots and on reconstruction fidelity, which rises from 15.97 dB to 21.74 dB. Under declared input, supervision and probe parity the protected confirmation stage reduces false repair for every detector to which it is attached, while a random forest supplied with the identical evidence reaches 4.14 slots, so the contribution is the staged protocol rather than the belief estimator. Zero-shot evaluation retains full detection with zero observed false repair on 5 unseen link models, on larger codebooks, on the unseen MNIST and Fashion-MNIST sources and in 6 of 7 unseen mismatch families and identifies receiver-side decoder drift as undetectable by the present observation model for every evaluated method. The control traffic share reaches 23.59 percent and is 12.62 percent lower than the common monitor value of 27.00 percent. These gains are achieved with 9.29 kbit of absolute control traffic and 0.393 mJ of radio energy per stream. The control-adjusted semantic resource efficiency is 6.203 source-equivalent bits per channel use, which remains below the best monitor because the useful task belief consumes additional radio resources. The task belief ablation reduces aggregate task accuracy to 83.9 percent and reduces overhead by 7.52 kbit, which exposes the preservation and communication tradeoff directly. The equal-budget Conditional CTM comparator also attains zero observed false repairs with shorter delay; lower energy and lower absolute signalling are obtained by the scalar monitors, so those metrics are not claimed as advantages of CoSMA DAI.

Future work will concentrate on five extensions. Decoder-side attestation will be added so that receiver task-head drift, which is invisible to the present fingerprint evidence, becomes observable. A harm-aware repair guard will be developed so that a confirmed displacement is repaired only when the induced functional outage justifies the codebook transfer. The task-preservation belief will be replaced by a compressed posterior or a low-rate summary statistic so that structured output spaces can be supported. Both learned beliefs will be evaluated under domain shift without retraining. The task belief will be extended from a top class decision to calibrated compressed posteriors for larger text, image and multimodal semantic tasks. Multiple interacting semantic links and measured radio traces will be examined. Hardware execution will be used to quantify inference latency, energy and codebook transfer cost. These extensions are required before deployment claims can be justified.

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Abbreviations

The following abbreviations are used in this manuscript:

BDIx	Extended Belief Desire Intention
CRC	Cyclic redundancy check
CTM	Conformal test martingale
CUSUM	Cumulative sum

DAI	Distributed artificial intelligence
ICML	International Conference on Machine Learning
PCA	Principal component analysis
TB	Task belief
SINR	Signal to interference plus noise ratio
WATCH	Weighted adaptive testing for change

Appendix A. Symbols and Notation

This appendix consolidates the symbols used in the system model, machine-learned belief formulation, sequential decision process and network efficiency accounting in Tables A1 and A2. Symbols that carry a method index use m and symbols that carry a time index use t .

Table A1. System, semantic representation and physical channel symbols.

Symbol	Definition	Symbol	Definition
t, T	Current slot and total number of evaluated slots	Δ	Duration of one communication slot
x_t	Source observation at slot t	$z_t = f_\theta(x_t)$	Learned semantic latent vector
d_z	Latent dimension	N_c	Number of semantic codewords
$\mathcal{C}^{\text{tx}}$	Transmitter semantic codebook	$\mathcal{C}_t^{\text{rx}}$	Receiver semantic codebook at slot t
c_k^{tx}	Transmitter codeword with index k	$c_{k,t}^{\text{rx}}$	Receiver codeword with index k at slot t
k_t	Transmitted semantic codeword index	q_k	Compact fingerprint associated with codeword k
R	Fixed orthonormal fingerprint projection	g_ψ	Receiver downstream task classifier
h_ω	Transmitter task belief classifier	π_t^{task}	Transmitter task posterior
$\hat{y}_t^B$	Compact transmitter task belief	y_t	Ground truth task label used only for evaluation
$\mathcal{K}_t^{\text{ack}}$	Last acknowledged common semantic state	$\mathcal{T}$	Downstream semantic task
v_t, a_t, χ_t	Version, acknowledgement state and alignment confidence	$\mathcal{O}_t^{\text{sem}}$	Functional semantic outage event
P_{tx}	Radio transmit power	h_t	Rician small scale fading coefficient
K	Rician factor	$L(d_t)$	Distance dependent path loss
d_t	Link distance	I_t	Interference power
N_0	Noise spectral density	B	Radio bandwidth
Γ_t, γ_t	Linear and decibel signal to interference plus noise ratio	p_b	Uncoded binary phase shift keying bit error probability
$p_b^{(3)}$	Residual bit error probability after threefold repetition	b_{ch}	Channel uses charged for one coded semantic payload

Table A2. Evidence, decision, statistical and network efficiency symbols.

Symbol	Definition	Symbol	Definition
s_t	Payload semantic consistency score	b_t	Channel quality or signal to interference plus noise ratio stratum
$m_{b,k}, r_{b,k}$	Reference median and robust scale for stratum b and codeword k	ϵ_r	Positive numerical floor for robust normalisation
z_t^{ctx}	Channel conditioned positive semantic excess	ρ	Evidence memory decay factor
G_t	Global accumulated evidence	$L_{j,t}$	Semantic class local accumulated evidence for class j

Table A2. Cont.

Symbol	Definition	Symbol	Definition
w_l	Local evidence weight	E_t	Fused passive evidence
s_t^{probe}	Protected probe semantic consistency score	P_t	Normalised protected probe evidence
$\mathbf{x}_t^B$	Causal feature vector for mismatch belief revision	B_t	Learned mismatch decision score
h_E	Persistent evidence gate	h_L	Semantic class local evidence gate
θ_B	Learned mismatch belief gate	c_E	Required number of consecutive physical slots
τ_s	Effective semantic drift onset	τ_{alarm}	Slot at which a repair is committed
D	Detection delay, $\tau_{\text{alarm}} - \tau_s$	FAR	False alarm rate on matched null streams
b_s	Source equivalent bits assigned to one task sample	S_m	Useful semantic side information transmitted by method m
C_m	Probe, repair and acknowledgement control traffic for method m	$R_{w,m}$	Task weighted source equivalent data rate
$\eta_{s,m}$	Bandwidth normalised semantic spectral efficiency	$\zeta_{s,m}$	Control adjusted semantic resource efficiency
$\rho_{c,m}$	Control traffic share	$E_{\text{tx},m}$	Modelled transmitter radio energy
P_{tx}/B	Energy per charged channel use under the disclosed accounting model	$\mathbb{I}[\cdot]$	Indicator function
Ω	Residual functional semantic outage rate after onset	$\mathcal{A}$	Set of accepted post-onset slots
τ_{cand}	Slot at which the passive gate opens protected confirmation	τ_{ack}	Slot at which codebook repair is acknowledged
$\hat{x}_t^{\text{rx}}$	Source reconstruction from the received semantic representation	d_x	Source observation dimension used for reconstruction fidelity
$\mathcal{I}_t$	Selected guarded intention	A_t^{pay}	Payload acceptance indicator, distinct from a_t

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